

B.TECH – FIRST YEAR | FIRST SEMESTER

Engineering Mathematics

Unit IV

Vector Differentiation

Topics: Scalar & Vector Functions · Del Operator (∇) · Gradient · Directional Derivative

Divergence · Curl · Vector Identities · Applications

40+ Ultra Step-by-Step Solved Examples

50 Exercise Problems · Complete Formula Sheet · Quick Revision Notes

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Section 1 – Introduction to Vector Differentiation

Vector Differentiation is one of the most powerful branches of calculus applied to engineering problems. It deals with the **differentiation of vector functions** — functions whose values are vectors (having both magnitude and direction) — and **scalar functions** defined over regions of space.

In classical (scalar) calculus, we differentiate a function like $f(x)$ which gives a single number. In vector calculus, we deal with functions like $F(x, y, z)$ which may return a scalar or a vector, and they live in 3-dimensional space.

Why Do Engineers Need Vector Differentiation?

Engineering Field	Application of Vector Differentiation
Fluid Mechanics	Velocity fields, flow rates, divergence for compressibility
Electromagnetics	Electric field (gradient of potential), curl for magnetic fields
Heat Transfer	Temperature gradient, direction of maximum heat flow
Structural Engineering	Stress and strain fields in 3D bodies
Robotics / Control	Position vectors, velocity, angular velocity (curl-related)

Coordinate System Reminder

We work in the **3-dimensional Cartesian coordinate system** with three mutually perpendicular axes: X, Y, and Z.

Every point in space is identified by coordinates **(x, y, z)**.

Three **unit vectors** (vectors of length 1) along these axes are:

- i** — unit vector along the positive X-axis: $i = \langle 1, 0, 0 \rangle$
- j** — unit vector along the positive Y-axis: $j = \langle 0, 1, 0 \rangle$
- k** — unit vector along the positive Z-axis: $k = \langle 0, 0, 1 \rangle$

Any vector **A** in 3D space is written as: $A = A_1i + A_2j + A_3k$

What is a Partial Derivative? (Essential Background)

Before studying vector differentiation, you must understand **partial derivatives**. When a function depends on multiple variables (e.g., $f(x, y, z)$), the partial derivative with respect to one variable treats all other variables as **constants**.

$\partial f / \partial x$ means: differentiate f with respect to x , keeping y and z constant. $\partial f / \partial y$ means: differentiate f with respect to y , keeping x and z constant. $\partial f / \partial z$ means: differentiate f with respect to z , keeping x and y constant. Example: If $f(x, y, z) = x^2y + yz^3 + 5xz$ $\partial f / \partial x = 2xy + 5z$ (treat y, z as constants) $\partial f / \partial y = x^2 + z^3$ (treat x, z as constants) $\partial f / \partial z = 3yz^2 + 5x$ (treat x, y as constants)

Section 2 – Scalar Point Functions

Definition: A **Scalar Point Function** is a function that assigns a *single real number (scalar)* to each point (x, y, z) in a region of space.

It is a rule: $\varphi : \text{Region in } \mathbb{R}^3 \rightarrow \mathbb{R}$ (i.e., from a 3D space point to a single number).

Notation: $\varphi(x, y, z)$ or $f(x, y, z)$

Examples of Scalar Point Functions

Scalar Function	Physical Meaning	At point (1,2,3)
$\varphi = x^2 + y^2 + z^2$	Square of distance from origin	$1+4+9 = 14$
$T(x,y,z) = 100 - 2x - 3y + z$	Temperature at a point	$100-2-6+3 = 95^\circ$
$p(x,y,z) = xyz$	Pressure distribution	6 units
$\varphi = 3x^2y - y^2z + xz^3$	Potential energy field	$3(1)(2)-4(3)+(1)(27)=6-12+27=21$

Scalar Field

When a scalar point function is defined over a region in space, it creates a **scalar field**. For example, the temperature at every point inside a room forms a temperature scalar field.

Key Note: The output of a scalar point function is always a single number — no direction involved. This is the fundamental difference from a vector point function.

Section 3 – Vector Point Functions

Definition: A **Vector Point Function** is a function that assigns a *vector* (having both magnitude and direction) to each point (x, y, z) in a region of space.

Notation: $\mathbf{F}(x, y, z) = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$

where F_1, F_2, F_3 are scalar functions of x, y, z (called *components*).

Examples of Vector Point Functions

Vector Function	Physical Meaning	At point (1,2,1)
$\mathbf{F} = x^2\mathbf{i} + xy\mathbf{j} + yz\mathbf{k}$	Force field in 3D	$\mathbf{F} = 1\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$
$\mathbf{V} = yz\mathbf{i} + xz\mathbf{j} + xy\mathbf{k}$	Velocity of a fluid	$\mathbf{V} = 2\mathbf{i} + \mathbf{j} + 2\mathbf{k}$
$\mathbf{E} = 2xi - 3yj + zk$	Electric field vector	$\mathbf{E} = 2\mathbf{i} - 6\mathbf{j} + \mathbf{k}$

Vector Field

A vector field assigns a vector at every point in space. Think of a flowing river — at each point on the surface, the water has a certain velocity (a vector) — that's a vector field.

Differentiation of a Vector Function (with respect to a scalar parameter)

If $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$, where t is time, then:

$\frac{d\mathbf{r}}{dt} = \left(\frac{dx}{dt}\right)\mathbf{i} + \left(\frac{dy}{dt}\right)\mathbf{j} + \left(\frac{dz}{dt}\right)\mathbf{k}$ This gives the velocity vector when $\mathbf{r}(t)$ is the position vector. $\frac{d^2\mathbf{r}}{dt^2}$ gives the acceleration vector.

Section 4 – The Del Operator (∇)

The Del Operator (written as ∇ , pronounced "nabla" or "del") is a *vector differential operator*. It is not a number — it is an *instruction* to differentiate.

It is defined in 3D Cartesian coordinates as:

$$\nabla = i(\partial/\partial x) + j(\partial/\partial y) + k(\partial/\partial z)$$
 Read as: "i times partial d by dx, PLUS j times partial d by dy, PLUS k times partial d by dz" In component form: $\nabla = (\partial/\partial x, \partial/\partial y, \partial/\partial z)$

Why is ∇ Called a Vector Operator?

The del operator behaves like a vector because it has three directional components (one for each axis). But it doesn't give a value by itself — it must be *applied* to either a scalar function or a vector function.

Operation	Symbol	Applied To	Result Type
Gradient	$\nabla\phi$	Scalar function ϕ	Vector
Divergence	$\nabla \cdot F$	Vector function F	Scalar
Curl	$\nabla \times F$	Vector function F	Vector
Laplacian	$\nabla^2\phi$	Scalar function ϕ	Scalar

Remember: ∇ alone is meaningless. It must always operate on something. The type of operation (gradient, divergence, curl) determines how ∇ interacts with the function.

Section 5 – Gradient of a Scalar Function

Definition: The **Gradient** of a scalar function $\phi(x, y, z)$ is obtained by applying the del operator ∇ to ϕ . The result is a *vector*.

$\text{grad } \phi = \nabla\phi = i(\partial\phi/\partial x) + j(\partial\phi/\partial y) + k(\partial\phi/\partial z)$ Expanded: $\nabla\phi = (\partial\phi/\partial x)i + (\partial\phi/\partial y)j + (\partial\phi/\partial z)k$ Each component is the rate of change of ϕ in that direction.

Step-by-Step Procedure to Find Gradient

Step	Action	Notes
1	Write down $\phi(x,y,z)$	Identify the scalar function
2	Compute $\partial\phi/\partial x$	Treat y, z as constants; differentiate w.r.t. x
3	Compute $\partial\phi/\partial y$	Treat x, z as constants; differentiate w.r.t. y
4	Compute $\partial\phi/\partial z$	Treat x, y as constants; differentiate w.r.t. z
5	Assemble $\nabla\phi$	$\nabla\phi = (\partial\phi/\partial x)i + (\partial\phi/\partial y)j + (\partial\phi/\partial z)k$
6	Evaluate at given point	Substitute x, y, z values if required

Properties of Gradient

- Linearity:** $\nabla(a\phi + b\psi) = a\nabla\phi + b\nabla\psi$ (a, b are constants)
- Product Rule:** $\nabla(\phi\psi) = \phi\nabla\psi + \psi\nabla\phi$
- Chain Rule:** $\nabla[f(\phi)] = f'(\phi) \cdot \nabla\phi$
- Gradient of constant = 0:** $\nabla(c) = 0$
- Gradient of r :** $\nabla(r) = r/r = \hat{r}$ (unit position vector), where $r = \sqrt{x^2+y^2+z^2}$

Section 6 – Physical Meaning of Gradient

The Gradient Points in the Direction of Maximum Increase

Imagine you are standing on a hilly terrain and the height of the terrain at each point (x,y) is given by a scalar function $\phi(x,y)$. The gradient $\nabla\phi$ at any point gives you the **direction and magnitude of the steepest ascent** — i.e., the direction you should walk to climb most steeply uphill.

Physical Interpretations:

- **Temperature field $T(x,y,z)$:** ∇T points toward the direction of maximum temperature increase. Heat flows in the direction of $-\nabla T$ (from hot to cold).
- **Electric potential $V(x,y,z)$:** Electric field $E = -\nabla V$. The gradient tells how steeply the potential changes.
- **Pressure field $p(x,y,z)$:** Fluid accelerates in the direction of $-\nabla p$ (from high to low pressure).

Gradient is Normal to Level Surfaces

A **level surface** (or equipotential surface) is a surface where $\phi(x,y,z) = \text{constant}$. For example, isothermal surfaces are where temperature = constant.

Key Theorem: The gradient $\nabla\phi$ is perpendicular (normal) to the level surface $\phi = c$ at every point on that surface.

This is why $\nabla\phi$ evaluated at a point on a surface gives the **normal vector** to that surface at that point — extremely useful in finding equations of tangent planes!

Magnitude of Gradient

$|\nabla\phi| = \sqrt{(\partial\phi/\partial x)^2 + (\partial\phi/\partial y)^2 + (\partial\phi/\partial z)^2}$ This magnitude tells you the RATE of maximum change of ϕ . The larger $|\nabla\phi|$, the steeper the function changes at that point.

Section 7 – Directional Derivative

Definition: The **Directional Derivative** of a scalar function ϕ in the direction of a vector $\mathbf{a}$ is the rate of change of ϕ as we move in that specific direction.

It tells us: "How fast does ϕ change per unit distance as I walk in the direction of $\mathbf{a}$?"

Directional Derivative of ϕ in direction of vector $\mathbf{a}$: $D_{\mathbf{a}} \phi = (\nabla \phi) \cdot \hat{\mathbf{a}}$ where $\hat{\mathbf{a}} = \mathbf{a} / |\mathbf{a}|$ is the UNIT VECTOR in the direction of $\mathbf{a}$. Expanded: $D_{\mathbf{a}} \phi = (\partial \phi / \partial x) (a_1 / |\mathbf{a}|) + (\partial \phi / \partial y) (a_2 / |\mathbf{a}|) + (\partial \phi / \partial z) (a_3 / |\mathbf{a}|)$

Why Must We Use a Unit Vector?

The directional derivative measures the rate of change *per unit distance*. If we used a non-unit vector, the result would be scaled by the vector's length, which is physically meaningless. The unit vector ensures we're measuring change per unit distance traveled.

Step-by-Step Procedure for Directional Derivative

Step	Action
1	Find gradient $\nabla \phi$ (using Section 5 method)
2	Evaluate $\nabla \phi$ at the given point (x_0, y_0, z_0)
3	Find the magnitude $ \mathbf{a} $ of the given direction vector $\mathbf{a}$
4	Compute unit vector $\hat{\mathbf{a}} = \mathbf{a} / \mathbf{a} $
5	Take dot product: $D_{\mathbf{a}} \phi = \nabla \phi \cdot \hat{\mathbf{a}}$

Key Facts About Directional Derivative

- Maximum directional derivative = $|\nabla \phi|$ (occurs in the direction of $\nabla \phi$)
- Minimum directional derivative = $-|\nabla \phi|$ (occurs in direction opposite to $\nabla \phi$)
- Directional derivative = 0 when the direction is perpendicular to $\nabla \phi$ (along the level surface)

Section 8 – Unit Normal Vector

Definition: The **Unit Normal Vector** to a surface $\phi(x,y,z) = c$ at a point P is the unit vector in the direction of $\nabla\phi$ at P.

Unit Normal Vector: $\hat{n} = \nabla\phi / |\nabla\phi|$ This vector is perpendicular (normal) to the surface at the given point.

Equation of Tangent Plane to a Surface

If the surface is $\phi(x,y,z) = c$, and $P = (x_0, y_0, z_0)$ is a point on the surface, then the tangent plane at P is:

$(\partial\phi/\partial x)_0(x - x_0) + (\partial\phi/\partial y)_0(y - y_0) + (\partial\phi/\partial z)_0(z - z_0) = 0$ Where the subscript 0 means evaluated at $P = (x_0, y_0, z_0)$.

Section 9 – Divergence of a Vector Field

Definition: The **Divergence** of a vector field $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$ is obtained by taking the *dot product* of the del operator ∇ with the vector $\mathbf{F}$.

The result is always a **scalar**.

$$\text{div } \mathbf{F} = \nabla \cdot \mathbf{F} = (\partial/\partial x, \partial/\partial y, \partial/\partial z) \cdot (F_1, F_2, F_3) \quad \nabla \cdot \mathbf{F} = \partial F_1/\partial x + \partial F_2/\partial y + \partial F_3/\partial z$$
 KEY: This is the sum of partial derivatives of each component with respect to its own variable.

Step-by-Step Procedure for Divergence

Step	Action
1	Identify F_1 (x-component), F_2 (y-component), F_3 (z-component)
2	Compute $\partial F_1/\partial x$ (differentiate F_1 w.r.t. x only)
3	Compute $\partial F_2/\partial y$ (differentiate F_2 w.r.t. y only)
4	Compute $\partial F_3/\partial z$ (differentiate F_3 w.r.t. z only)
5	Add all three: $\nabla \cdot \mathbf{F} = \partial F_1/\partial x + \partial F_2/\partial y + \partial F_3/\partial z$
6	If a specific point is given, substitute x, y, z values

Properties of Divergence

- Linearity:** $\nabla \cdot (\mathbf{F} + \mathbf{G}) = \nabla \cdot \mathbf{F} + \nabla \cdot \mathbf{G}$
- Scalar multiplication:** $\nabla \cdot (c\mathbf{F}) = c(\nabla \cdot \mathbf{F})$ where c is a constant
- Product rule:** $\nabla \cdot (\phi\mathbf{F}) = \phi(\nabla \cdot \mathbf{F}) + (\nabla\phi) \cdot \mathbf{F}$
- Solenoidal field:** If $\nabla \cdot \mathbf{F} = 0$, then $\mathbf{F}$ is called a solenoidal (or divergence-free) vector field

Section 10 – Physical Interpretation of Divergence

Divergence Measures "Source" or "Sink"

Think of a vector field as the velocity field of a fluid. Divergence at a point measures how much fluid is **flowing outward** from (or inward into) an infinitesimally small volume around that point.

- $\nabla \cdot \mathbf{F} > 0$ at $\mathbf{P}$: More fluid is flowing OUT than in at $\mathbf{P}$. $\mathbf{P}$ is a *source* (fluid is being created or expanding there).
- $\nabla \cdot \mathbf{F} < 0$ at $\mathbf{P}$: More fluid is flowing IN than out at $\mathbf{P}$. $\mathbf{P}$ is a *sink* (fluid is being absorbed or compressed there).
- $\nabla \cdot \mathbf{F} = 0$ at $\mathbf{P}$: Fluid entering = fluid leaving. The flow is *incompressible* at $\mathbf{P}$ (no source, no sink). Such a field is called *solenoidal*.

Examples of Physical Interpretation

Field	$\nabla \cdot \mathbf{F}$ Meaning
Velocity field of fluid	Positive = compression zone; Negative = expansion zone; Zero = incompressible
Electric field $\mathbf{E}$	$\nabla \cdot \mathbf{E} = \rho/\epsilon_0$ (Gauss's Law: div of $\mathbf{E}$ = charge density / permittivity)
Heat flux $\mathbf{q}$	$\nabla \cdot \mathbf{q}$ measures heat generated or absorbed per unit volume

Section 11 – Curl of a Vector Field

Definition: The **Curl** of a vector field $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$ is obtained by taking the *cross product* of the del operator ∇ with the vector $\mathbf{F}$.

The result is always a **vector**.

$$\text{curl } \mathbf{F} = \nabla \times \mathbf{F} \text{ Using cross product definition: } \nabla \times \mathbf{F} = \mathbf{i}(\partial F_3/\partial y - \partial F_2/\partial z) - \mathbf{j}(\partial F_3/\partial x - \partial F_1/\partial z) + \mathbf{k}(\partial F_2/\partial x - \partial F_1/\partial y)$$

Physical Meaning of Curl

Curl measures the **rotational tendency** (or swirling nature) of a vector field at a point.

- If you placed a tiny paddle wheel in a fluid with velocity field $\mathbf{F}$, the curl tells you how fast and in what direction the paddle wheel would spin.
- Large curl = high rotation. Zero curl = no rotation at that point (irrotational field).

- $\nabla \times \mathbf{F} = \mathbf{0}$: The field is called *irrotational*. No rotation at that point.
- $|\nabla \times \mathbf{F}|$: Magnitude of rotation (angular velocity of the paddle wheel is half of this).
- **Direction of $\nabla \times \mathbf{F}$** : The axis about which the fluid rotates (right-hand rule).

Section 12 – Determinant Method for Curl

The most systematic way to compute curl is using the **3×3 symbolic determinant** method. This method ensures no terms are skipped.

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$$

Expanding the Determinant – Step-by-Step

We expand along the first row using cofactor expansion:

$\nabla \times \mathbf{F} = \mathbf{i} \cdot M_{11} - \mathbf{j} \cdot M_{12} + \mathbf{k} \cdot M_{13}$ Where the 2×2 minors are: $M_{11} = \begin{vmatrix} \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_2 & F_3 \end{vmatrix} = (\frac{\partial F_3}{\partial y}) - (\frac{\partial F_2}{\partial z})$ $M_{12} = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial z} \\ F_1 & F_3 \end{vmatrix} = (\frac{\partial F_3}{\partial x}) - (\frac{\partial F_1}{\partial z})$ $M_{13} = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \\ F_1 & F_2 \end{vmatrix} = (\frac{\partial F_2}{\partial x}) - (\frac{\partial F_1}{\partial y})$ Therefore: $\nabla \times \mathbf{F} = \mathbf{i}(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}) - \mathbf{j}(\frac{\partial F_3}{\partial x} - \frac{\partial F_1}{\partial z}) + \mathbf{k}(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y})$

Memory Aid for Curl Components

Component	Formula	Sign
i-component	$\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}$	+
j-component	$\frac{\partial F_3}{\partial x} - \frac{\partial F_1}{\partial z}$	– (negative!)
k-component	$\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}$	+

COMMON MISTAKE: Students often forget the negative sign on the j-component! The cofactor expansion of a 3×3 determinant along the first row has signs: +, –, +. The j-term always gets a minus sign.

Section 13 – Vector Identities

Important Vector Identities Involving ∇

#	Identity	Notes
1	$\nabla(\varphi + \psi) = \nabla\varphi + \nabla\psi$	Gradient is linear
2	$\nabla(\varphi\psi) = \varphi\nabla\psi + \psi\nabla\varphi$	Product rule for gradient
3	$\nabla \cdot (\mathbf{F} + \mathbf{G}) = \nabla \cdot \mathbf{F} + \nabla \cdot \mathbf{G}$	Divergence is linear
4	$\nabla \cdot (\varphi\mathbf{F}) = \varphi(\nabla \cdot \mathbf{F}) + (\nabla\varphi) \cdot \mathbf{F}$	Product rule for divergence
5	$\nabla \times (\mathbf{F} + \mathbf{G}) = \nabla \times \mathbf{F} + \nabla \times \mathbf{G}$	Curl is linear
6	$\nabla \times (\varphi\mathbf{F}) = \varphi(\nabla \times \mathbf{F}) + (\nabla\varphi) \times \mathbf{F}$	Product rule for curl
7	$\nabla \cdot (\nabla \times \mathbf{F}) = 0$	Div of curl = always ZERO
8	$\nabla \times (\nabla\varphi) = 0$	Curl of gradient = always ZERO
9	$\nabla \times (\nabla \times \mathbf{F}) = \nabla(\nabla \cdot \mathbf{F}) - \nabla^2\mathbf{F}$	Curl of curl formula
10	$\nabla \cdot (\mathbf{F} \times \mathbf{G}) = \mathbf{G} \cdot (\nabla \times \mathbf{F}) - \mathbf{F} \cdot (\nabla \times \mathbf{G})$	Useful identity

Explanation of Identity 7: $\nabla \cdot (\nabla \times \mathbf{F}) = 0$

If you take any vector field $\mathbf{F}$, compute its curl $\nabla \times \mathbf{F}$ (which is another vector field), and then compute the divergence of that result — you always get zero. This is a fundamental theorem of vector calculus and is used in electromagnetism (Maxwell's equations).

Explanation of Identity 8: $\nabla \times (\nabla\varphi) = 0$

If you take the gradient of any scalar function φ (which gives a vector field), and then compute the curl of that gradient — you always get the zero vector. This means gradient fields are always irrotational. This is why conservative force fields (like gravity) have zero curl.

PROOF OF IMPORTANT VECTOR IDENTITIES INVOLVING THE DEL OPERATOR (∇)

Complete Step-by-Step Proofs from First Principles • Exam-Oriented Format

7 Identities

All Steps Shown

Component Form

Product Rule Explicit

Fundamental Definitions (Used Throughout All Proofs)

Scalar functions: $\phi(x, y, z)$ and $\psi(x, y, z)$ — real-valued, continuously differentiable.

Vector function:

$$\mathbf{A} = A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k}$$

where A_x, A_y, A_z are scalar functions of (x, y, z) .

The Del Operator:

$$\nabla = \mathbf{i} \partial/\partial x + \mathbf{j} \partial/\partial y + \mathbf{k} \partial/\partial z$$

Del is a *vector differential operator*. It is **not** a vector by itself — it only makes sense when acting on a scalar or vector field. When it acts on a scalar, it produces a **gradient** (vector). When dotted with a vector, it produces a **divergence** (scalar). When crossed with a vector, it produces a **curl** (vector).

Product Rule for derivatives (used extensively):

$$\partial(fg)/\partial x = f \partial g/\partial x + g \partial f/\partial x$$

Gradient of a Sum

$$\nabla (\varphi + \psi) = \nabla \varphi + \nabla \psi$$

1 DEFINE VARIABLES

Let $\varphi(x, y, z)$ and $\psi(x, y, z)$ be scalar functions, each continuously differentiable with respect to all three variables x , y , and z .

We want to prove that the gradient of their sum equals the sum of their gradients.

2 WRITE THE DEL OPERATOR EXPLICITLY

$$\nabla = \mathbf{i} \partial/\partial x + \mathbf{j} \partial/\partial y + \mathbf{k} \partial/\partial z$$

When ∇ acts on a scalar function f , it gives the **gradient**:

$$\nabla f = \mathbf{i} \partial f/\partial x + \mathbf{j} \partial f/\partial y + \mathbf{k} \partial f/\partial z$$

3 EXPAND THE LEFT-HAND SIDE (LHS)

Apply ∇ to the sum $(\varphi + \psi)$:

$$\begin{aligned} \text{LHS} &= \nabla (\varphi + \psi) \\ &= \mathbf{i} \partial (\varphi + \psi) / \partial x + \mathbf{j} \partial (\varphi + \psi) / \partial y + \mathbf{k} \partial (\varphi + \psi) / \partial z \end{aligned}$$

4 APPLY PARTIAL DIFFERENTIATION — ONE VARIABLE AT A TIME

Differentiation is a linear operation, so we can separate derivatives of sums:

Partial Derivative	Expansion
$\partial(\varphi+\psi)/\partial x$	$= \partial\varphi/\partial x + \partial\psi/\partial x$ [by linearity of differentiation]
$\partial(\varphi+\psi)/\partial y$	$= \partial\varphi/\partial y + \partial\psi/\partial y$

Partial Derivative	Expansion
$\partial(\varphi+\psi)/\partial z$	$= \partial\varphi/\partial z + \partial\psi/\partial z$

Substituting these back into the LHS expression:

$$\text{LHS} = \mathbf{i} (\partial\varphi/\partial x + \partial\psi/\partial x) + \mathbf{j} (\partial\varphi/\partial y + \partial\psi/\partial y) + \mathbf{k} (\partial\varphi/\partial z + \partial\psi/\partial z)$$

5 EXPAND THE RIGHT-HAND SIDE (RHS) SEPARATELY

Now compute $\nabla\varphi$ and $\nabla\psi$ individually:

$$\nabla\varphi = \mathbf{i} \partial\varphi/\partial x + \mathbf{j} \partial\varphi/\partial y + \mathbf{k} \partial\varphi/\partial z$$

$$\nabla\psi = \mathbf{i} \partial\psi/\partial x + \mathbf{j} \partial\psi/\partial y + \mathbf{k} \partial\psi/\partial z$$

Adding them:

$$\begin{aligned} \text{RHS} &= \nabla\varphi + \nabla\psi \\ &= \mathbf{i} (\partial\varphi/\partial x + \partial\psi/\partial x) + \mathbf{j} (\partial\varphi/\partial y + \partial\psi/\partial y) + \mathbf{k} (\partial\varphi/\partial z + \partial\psi/\partial z) \end{aligned}$$

6 COMPARE LHS AND RHS COMPONENT-WISE

Component	LHS	RHS	Equal?
i-component	$\partial\varphi/\partial x + \partial\psi/\partial x$	$\partial\varphi/\partial x + \partial\psi/\partial x$	✓ Yes
j-component	$\partial\varphi/\partial y + \partial\psi/\partial y$	$\partial\varphi/\partial y + \partial\psi/\partial y$	✓ Yes
k-component	$\partial\varphi/\partial z + \partial\psi/\partial z$	$\partial\varphi/\partial z + \partial\psi/\partial z$	✓ Yes

All three components match. Therefore $\text{LHS} = \text{RHS}$.

7 FINAL CONCLUSION

✓ IDENTITY PROVED

$$\nabla(\varphi + \psi) = \nabla\varphi + \nabla\psi$$

Why is this important?

This identity confirms that the gradient operator ∇ is *linear* — exactly like ordinary derivatives. This means we can always break complex scalar fields into simpler parts, compute their gradients separately, and add up the results. In physics, this is the backbone of the principle of superposition applied to potential fields (e.g., the electric potential of multiple charges).

✓ IDENTITY 1 PROVED — $\nabla(\phi + \psi) = \nabla\phi + \nabla\psi$

Divergence of a Scalar–Vector Product

$$\nabla \cdot (\varphi \mathbf{A}) = \varphi (\nabla \cdot \mathbf{A}) + (\nabla \varphi) \cdot \mathbf{A}$$

1

DEFINE VARIABLES

Let $\varphi(x,y,z)$ be a scalar function. Let $\mathbf{A}$ be a vector field:

$$\mathbf{A} = A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k}$$

The product $\varphi \mathbf{A}$ is a vector whose each component is multiplied by φ :

$$\varphi \mathbf{A} = (\varphi A_x) \mathbf{i} + (\varphi A_y) \mathbf{j} + (\varphi A_z) \mathbf{k}$$

2

WRITE DIVERGENCE USING THE DEL OPERATOR

Divergence of a vector $\mathbf{F} = F_x \mathbf{i} + F_y \mathbf{j} + F_z \mathbf{k}$ is:

$$\nabla \cdot \mathbf{F} = \partial F_x / \partial x + \partial F_y / \partial y + \partial F_z / \partial z$$

Here, our vector is $\varphi \mathbf{A}$, so $F_x = \varphi A_x$, $F_y = \varphi A_y$, $F_z = \varphi A_z$.

3

EXPAND THE LHS COMPLETELY

$$\begin{aligned} \text{LHS} &= \nabla \cdot (\varphi \mathbf{A}) \\ &= \partial (\varphi A_x) / \partial x + \partial (\varphi A_y) / \partial y + \partial (\varphi A_z) / \partial z \end{aligned}$$

4

APPLY THE PRODUCT RULE TO EACH TERM

For each term, apply: $\partial(\varphi f) / \partial x = \varphi \partial f / \partial x + f \partial \varphi / \partial x$

Term	Product Rule Expansion
$\partial(\varphi A_x) / \partial x$	$= \varphi \partial A_x / \partial x + A_x \partial \varphi / \partial x$
$\partial(\varphi A_y) / \partial y$	$= \varphi \partial A_y / \partial y + A_y \partial \varphi / \partial y$

Term	Product Rule Expansion
$\partial(\phi A_z)/\partial z$	$= \phi \partial A_z/\partial z + A_z \partial \phi/\partial z$

Substituting all three back into the LHS sum:

$$\begin{aligned} \text{LHS} &= \phi \partial A_x/\partial x + A_x \partial \phi/\partial x \\ &\quad + \phi \partial A_y/\partial y + A_y \partial \phi/\partial y \\ &\quad + \phi \partial A_z/\partial z + A_z \partial \phi/\partial z \end{aligned}$$

Regroup by collecting ϕ terms together and the $A \cdot \nabla \phi$ terms together:

$$\begin{aligned} \text{LHS} &= \phi (\partial A_x/\partial x + \partial A_y/\partial y + \partial A_z/\partial z) \\ &\quad + (A_x \partial \phi/\partial x + A_y \partial \phi/\partial y + A_z \partial \phi/\partial z) \end{aligned}$$

5

EXPAND THE RHS SEPARATELY

Term 1: $\phi(\nabla \cdot \mathbf{A})$

$$\begin{aligned} \nabla \cdot \mathbf{A} &= \partial A_x/\partial x + \partial A_y/\partial y + \partial A_z/\partial z \\ \Rightarrow \phi(\nabla \cdot \mathbf{A}) &= \phi(\partial A_x/\partial x + \partial A_y/\partial y + \partial A_z/\partial z) \end{aligned}$$

Term 2: $(\nabla \phi) \cdot \mathbf{A}$

$$\begin{aligned} \nabla \phi &= (\partial \phi/\partial x) \mathbf{i} + (\partial \phi/\partial y) \mathbf{j} + (\partial \phi/\partial z) \mathbf{k} \\ (\nabla \phi) \cdot \mathbf{A} &= \partial \phi/\partial x \cdot A_x + \partial \phi/\partial y \cdot A_y + \partial \phi/\partial z \cdot A_z \end{aligned}$$

Total RHS:

$$\begin{aligned} \text{RHS} &= \phi (\partial A_x/\partial x + \partial A_y/\partial y + \partial A_z/\partial z) \\ &\quad + (A_x \partial \phi/\partial x + A_y \partial \phi/\partial y + A_z \partial \phi/\partial z) \end{aligned}$$

6

COMPARE LHS AND RHS COMPONENT-WISE

Group	LHS	RHS	Equal?
Group 1 ($\phi \times \text{div } \mathbf{A}$)	$\phi (\partial A_x/\partial x + \partial A_y/\partial y + \partial A_z/\partial z)$	$\phi (\nabla \cdot \mathbf{A})$	✓ Yes

Group	LHS	RHS	Equal?
Group 2 ($\mathbf{A} \cdot \text{grad } \varphi$)	$A_x \partial \varphi / \partial x + A_y \partial \varphi / \partial y + A_z \partial \varphi / \partial z$	$(\nabla \varphi) \cdot \mathbf{A}$	✓ Yes

LHS = RHS identically.

7

FINAL CONCLUSION

✓ IDENTITY PROVED

$$\nabla \cdot (\varphi \mathbf{A}) = \varphi (\nabla \cdot \mathbf{A}) + (\nabla \varphi) \cdot \mathbf{A}$$

8

CONCEPT INSIGHT

Physical Meaning: This identity is the vector analog of the scalar product rule. In fluid dynamics, if $\mathbf{A}$ is velocity and φ is density, then $\nabla \cdot (\varphi \mathbf{A})$ gives the total divergence of mass flux. This identity tells us that this divergence has two sources: one due to the spatial variation of the velocity field ($\varphi \nabla \cdot \mathbf{A}$) and another due to the spatial variation of density $((\nabla \varphi) \cdot \mathbf{A})$.

✓ IDENTITY 2 PROVED — $\nabla \cdot (\varphi \mathbf{A}) = \varphi (\nabla \cdot \mathbf{A}) + (\nabla \varphi) \cdot \mathbf{A}$

Curl of a Scalar–Vector Product

$$\nabla \times (\varphi \mathbf{A}) = \varphi (\nabla \times \mathbf{A}) + (\nabla \varphi) \times \mathbf{A}$$

1

DEFINE VARIABLES

Let $\varphi(x,y,z)$ be a scalar. Let $\mathbf{A} = A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k}$.

Then: $\varphi \mathbf{A} = (\varphi A_x) \mathbf{i} + (\varphi A_y) \mathbf{j} + (\varphi A_z) \mathbf{k}$.

2

WRITE THE CURL USING THE DETERMINANT FORM

The curl of any vector $\mathbf{F} = (F_x, F_y, F_z)$ is given by:

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ F_x & F_y & F_z \end{vmatrix}$$

Here $F_x = \varphi A_x, F_y = \varphi A_y, F_z = \varphi A_z$. Expanding along the first row:

$$\begin{aligned} \nabla \times (\varphi \mathbf{A}) &= \mathbf{i} [\partial (\varphi A_z) / \partial y - \partial (\varphi A_y) / \partial z] \\ &\quad - \mathbf{j} [\partial (\varphi A_z) / \partial x - \partial (\varphi A_x) / \partial z] \\ &\quad + \mathbf{k} [\partial (\varphi A_y) / \partial x - \partial (\varphi A_x) / \partial y] \end{aligned}$$

3

APPLY THE PRODUCT RULE TO EVERY TERM

Applying $\partial(fg)/\partial x = f \partial g/\partial x + g \partial f/\partial x$ to each partial derivative:

Derivative	Product Rule Expansion
$\partial(\varphi A_z)/\partial y$	$= \varphi \partial A_z / \partial y + A_z \partial \varphi / \partial y$
$\partial(\varphi A_y)/\partial z$	$= \varphi \partial A_y / \partial z + A_y \partial \varphi / \partial z$
$\partial(\varphi A_z)/\partial x$	$= \varphi \partial A_z / \partial x + A_z \partial \varphi / \partial x$
$\partial(\varphi A_x)/\partial z$	$= \varphi \partial A_x / \partial z + A_x \partial \varphi / \partial z$

Derivative	Product Rule Expansion
$\partial(\phi A_y)/\partial x$	$= \phi \partial A_y / \partial x + A_y \partial \phi / \partial x$
$\partial(\phi A_x)/\partial y$	$= \phi \partial A_x / \partial y + A_x \partial \phi / \partial y$

4 SUBSTITUTE BACK INTO EACH COMPONENT

i-component:

$$\begin{aligned}
 & [\partial(\phi A_z) / \partial y - \partial(\phi A_y) / \partial z] \\
 &= (\phi \partial A_z / \partial y + A_z \partial \phi / \partial y) - (\phi \partial A_y / \partial z + A_y \partial \phi / \partial z) \\
 &= \phi (\partial A_z / \partial y - \partial A_y / \partial z) + (A_z \partial \phi / \partial y - A_y \partial \phi / \partial z)
 \end{aligned}$$

j-component (note the negative sign from cofactor expansion):

$$\begin{aligned}
 & -[\partial(\phi A_z) / \partial x - \partial(\phi A_x) / \partial z] \\
 &= -[(\phi \partial A_z / \partial x + A_z \partial \phi / \partial x) - (\phi \partial A_x / \partial z + A_x \partial \phi / \partial z)] \\
 &= \phi (\partial A_x / \partial z - \partial A_z / \partial x) + (A_x \partial \phi / \partial z - A_z \partial \phi / \partial x)
 \end{aligned}$$

k-component:

$$\begin{aligned}
 & [\partial(\phi A_y) / \partial x - \partial(\phi A_x) / \partial y] \\
 &= (\phi \partial A_y / \partial x + A_y \partial \phi / \partial x) - (\phi \partial A_x / \partial y + A_x \partial \phi / \partial y) \\
 &= \phi (\partial A_y / \partial x - \partial A_x / \partial y) + (A_y \partial \phi / \partial x - A_x \partial \phi / \partial y)
 \end{aligned}$$

5 COLLECT AND REGROUP — EXPAND RHS

Collecting all $\phi(\dots)$ terms gives:

$$\begin{aligned}
 & \phi \cdot [\mathbf{i}(\partial A_z / \partial y - \partial A_y / \partial z) + \mathbf{j}(\partial A_x / \partial z - \partial A_z / \partial x) + \mathbf{k}(\partial A_y / \partial x - \partial A_x / \partial y)] \\
 &= \phi (\nabla \times \mathbf{A})
 \end{aligned}$$

Collecting the cross-product terms gives:

$$\mathbf{i}(A_z \partial \phi / \partial y - A_y \partial \phi / \partial z) + \mathbf{j}(A_x \partial \phi / \partial z - A_z \partial \phi / \partial x) + \mathbf{k}(A_y \partial \phi / \partial x - A_x \partial \phi / \partial y)$$

This is exactly the cross product $(\nabla\phi) \times \mathbf{A}$. To verify, compute the determinant:

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial\phi/\partial x & \partial\phi/\partial y & \partial\phi/\partial z \\ A_x & A_y & A_z \end{vmatrix}$$

$$\begin{aligned} &= \mathbf{i} (A_z \partial\phi/\partial y - A_y \partial\phi/\partial z) - \mathbf{j} (A_z \partial\phi/\partial x - A_x \partial\phi/\partial z) + \mathbf{k} (A_y \partial\phi/\partial x - A_x \partial\phi/\partial y) \\ &= (\nabla\phi) \times \mathbf{A} \quad \checkmark \end{aligned}$$

6 FINAL COMPARISON

$$\text{LHS} = \nabla \times (\phi \mathbf{A}) = \phi (\nabla \times \mathbf{A}) + (\nabla \phi) \times \mathbf{A} = \text{RHS} \quad \checkmark$$

7 FINAL CONCLUSION

✓ IDENTITY PROVED

$$\nabla \times (\phi \mathbf{A}) = \phi (\nabla \times \mathbf{A}) + (\nabla \phi) \times \mathbf{A}$$

8 CONCEPT INSIGHT

Physical Meaning: This identity is the curl-analog of the product rule. In electromagnetism, if $\mathbf{A}$ is the magnetic vector potential and ϕ is a scalar, this tells us how the rotation (curl) of the scaled field $\phi \mathbf{A}$ decomposes. The term $(\nabla \phi) \times \mathbf{A}$ captures the contribution from spatial variation of the scalar, while $\phi (\nabla \times \mathbf{A})$ captures the rotation of the original vector field.

✓ IDENTITY 3 PROVED — $\nabla \times (\phi \mathbf{A}) = \phi (\nabla \times \mathbf{A}) + (\nabla \phi) \times \mathbf{A}$

Divergence of a Curl is Zero

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0$$

1 DEFINE VARIABLES

Let $\mathbf{A} = A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k}$ where A_x, A_y, A_z are twice continuously differentiable scalar functions of (x, y, z) .

2 COMPUTE THE CURL OF A — FULL DETERMINANT EXPANSION

$$\nabla \times \mathbf{A} =$$

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ A_x & A_y & A_z \end{vmatrix}$$

Expanding along Row 1 — cofactor by cofactor:

i-cofactor (minor from columns 2&3, rows 2&3):

$$+\mathbf{i} \cdot [(\partial/\partial y)(A_z) - (\partial/\partial z)(A_y)] = \mathbf{i}(\partial A_z/\partial y - \partial A_y/\partial z)$$

j-cofactor (minor from columns 1&3, rows 2&3 — sign is negative):

$$-\mathbf{j} \cdot [(\partial/\partial x)(A_z) - (\partial/\partial z)(A_x)] = \mathbf{j}(\partial A_x/\partial z - \partial A_z/\partial x)$$

k-cofactor (minor from columns 1&2, rows 2&3):

$$+\mathbf{k} \cdot [(\partial/\partial x)(A_y) - (\partial/\partial y)(A_x)] = \mathbf{k}(\partial A_y/\partial x - \partial A_x/\partial y)$$

So the full curl is:

$$\nabla \times \mathbf{A} = \mathbf{i}(\partial A_z/\partial y - \partial A_y/\partial z) + \mathbf{j}(\partial A_x/\partial z - \partial A_z/\partial x) + \mathbf{k}(\partial A_y/\partial x - \partial A_x/\partial y)$$

TAKE DIVERGENCE OF THE CURL

Let $\mathbf{C} = \nabla \times \mathbf{A} = C_x \mathbf{i} + C_y \mathbf{j} + C_z \mathbf{k}$, where:

Component	Expression
C_x	$\partial A_z / \partial y - \partial A_y / \partial z$
C_y	$\partial A_x / \partial z - \partial A_z / \partial x$
C_z	$\partial A_y / \partial x - \partial A_x / \partial y$

Now compute divergence:

$$\nabla \cdot \mathbf{C} = \partial C_x / \partial x + \partial C_y / \partial y + \partial C_z / \partial z$$

4 COMPUTE EACH PARTIAL DERIVATIVE EXPLICITLY

Derivative	Full Expansion
$\partial C_x / \partial x$	$\partial^2 A_z / \partial x \partial y - \partial^2 A_y / \partial x \partial z$
$\partial C_y / \partial y$	$\partial^2 A_x / \partial y \partial z - \partial^2 A_z / \partial y \partial x$
$\partial C_z / \partial z$	$\partial^2 A_y / \partial z \partial x - \partial^2 A_x / \partial z \partial y$

Adding all three:

$$\begin{aligned} &\nabla \cdot (\nabla \times \mathbf{A}) \\ &= (\partial^2 A_z / \partial x \partial y - \partial^2 A_y / \partial x \partial z) \\ &\quad + (\partial^2 A_x / \partial y \partial z - \partial^2 A_z / \partial z \partial y) \\ &\quad + (\partial^2 A_y / \partial z \partial x - \partial^2 A_x / \partial x \partial z) \end{aligned}$$

5 APPLY CLAIRAUT'S THEOREM (EQUALITY OF MIXED PARTIALS)

Clairaut's Theorem: For any twice continuously differentiable function f , mixed partial derivatives are equal: $\partial^2 f / \partial x \partial y = \partial^2 f / \partial y \partial x$

Applying this theorem to each pair:

Pair	By Clairaut's Theorem	Difference
$\partial^2 A_z / \partial x \partial y$ vs $\partial^2 A_z / \partial y \partial x$	They are equal	$= 0$
$\partial^2 A_x / \partial y \partial z$ vs $\partial^2 A_x / \partial z \partial y$	They are equal	$= 0$
$\partial^2 A_y / \partial z \partial x$ vs $\partial^2 A_y / \partial x \partial z$	They are equal	$= 0$

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0 + 0 + 0 = 0$$

7 FINAL CONCLUSION

✓ IDENTITY PROVED

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0 \quad (\text{for any twice-differentiable vector field } \mathbf{A})$$

8 CONCEPT INSIGHT

Physical Meaning: In electromagnetism, this identity guarantees that the divergence of the magnetic field $\mathbf{B} = \nabla \times \mathbf{A}$ is always zero — this is one of Maxwell's equations: $\nabla \cdot \mathbf{B} = 0$, which means there are no magnetic monopoles. Any field that is expressible as a curl automatically has zero divergence.

✓ IDENTITY 4 PROVED — $\nabla \cdot (\nabla \times \mathbf{A}) = 0$

Curl of a Gradient is Zero

$$\nabla \times (\nabla \phi) = 0$$

1 DEFINE VARIABLES

Let $\phi(x,y,z)$ be a twice continuously differentiable scalar function. We first compute $\nabla\phi$, then take its curl.

2 COMPUTE THE GRADIENT OF Φ

$$\nabla\phi = (\partial\phi/\partial x)\mathbf{i} + (\partial\phi/\partial y)\mathbf{j} + (\partial\phi/\partial z)\mathbf{k}$$

Let us call $P = \partial\phi/\partial x$, $Q = \partial\phi/\partial y$, $R = \partial\phi/\partial z$ so:

$$\nabla\phi = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$$

3 EXPAND THE CURL USING THE FULL DETERMINANT

$$\nabla \times (\nabla \phi) =$$

$\mathbf{i}$	$\mathbf{j}$	$\mathbf{k}$
$\partial/\partial x$	$\partial/\partial y$	$\partial/\partial z$
$\partial\phi/\partial x$	$\partial\phi/\partial y$	$\partial\phi/\partial z$

Expanding row 1 cofactor by cofactor:

i-component:

$$\begin{aligned} & \mathbf{i} [\partial/\partial y (\partial\phi/\partial z) - \partial/\partial z (\partial\phi/\partial y)] \\ &= \mathbf{i} [\partial^2\phi/\partial y\partial z - \partial^2\phi/\partial z\partial y] \end{aligned}$$

j-component (negative sign):

$$\begin{aligned} & -\mathbf{j} [\partial/\partial x (\partial\phi/\partial z) - \partial/\partial z (\partial\phi/\partial x)] \\ &= -\mathbf{j} [\partial^2\phi/\partial x\partial z - \partial^2\phi/\partial z\partial x] \end{aligned}$$

k-component:

$$\begin{aligned} &+ \mathbf{k} \left[\frac{\partial}{\partial x} \left(\frac{\partial \phi}{\partial y} \right) - \frac{\partial}{\partial y} \left(\frac{\partial \phi}{\partial x} \right) \right] \\ &= \mathbf{k} \left[\frac{\partial^2 \phi}{\partial x \partial y} - \frac{\partial^2 \phi}{\partial y \partial x} \right] \end{aligned}$$

4 APPLY CLAIRAUT'S THEOREM TO CANCEL ALL TERMS

Clairaut's Theorem: $\frac{\partial^2 \phi}{\partial y \partial z} = \frac{\partial^2 \phi}{\partial z \partial y}$ (and similarly for all mixed partials of ϕ)

Component	Expression	By Clairaut	Result
i	$\frac{\partial^2 \phi}{\partial y \partial z} - \frac{\partial^2 \phi}{\partial z \partial y}$	Both terms equal	$= 0$
j	$\frac{\partial^2 \phi}{\partial x \partial z} - \frac{\partial^2 \phi}{\partial z \partial x}$	Both terms equal	$= 0$
k	$\frac{\partial^2 \phi}{\partial x \partial y} - \frac{\partial^2 \phi}{\partial y \partial x}$	Both terms equal	$= 0$

$$\nabla \times (\nabla \phi) = 0\mathbf{i} + 0\mathbf{j} + 0\mathbf{k} = \mathbf{0}$$

7 FINAL CONCLUSION

✓ **IDENTITY PROVED**

$$\nabla \times (\nabla \phi) = \mathbf{0} \quad (\text{for any twice-differentiable scalar } \phi)$$

8 CONCEPT INSIGHT

Physical Meaning & Importance: This identity tells us that *conservative vector fields are irrotational*. If a force $\mathbf{F} = -\nabla \phi$ (e.g., gravity, electrostatics), then $\nabla \times \mathbf{F} = \mathbf{0}$, meaning it has no rotational component. This is the mathematical basis for the concept of potential energy. Conversely, if $\nabla \times \mathbf{F} \neq \mathbf{0}$, then $\mathbf{F}$ cannot be written as a gradient — it has no scalar potential.

✓ **IDENTITY 5 PROVED — $\nabla \times (\nabla \phi) = \mathbf{0}$**

Curl of Curl (Vector Triple Product Identity)

$$\nabla \times (\nabla \times \mathbf{A}) = \nabla (\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$$

1 DEFINE VARIABLES AND STRATEGY

Let $\mathbf{A} = A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k}$. We will compute the curl of curl by working component-by-component, and show that it equals the gradient of divergence minus the vector Laplacian.

The vector Laplacian is defined as:

$$\nabla^2 \mathbf{A} = (\nabla^2 A_x) \mathbf{i} + (\nabla^2 A_y) \mathbf{j} + (\nabla^2 A_z) \mathbf{k}$$

2 COMPUTE THE CURL OF A — FULL EXPANSION

$$\nabla \times \mathbf{A} = \mathbf{i} (\partial A_z / \partial y - \partial A_y / \partial z) + \mathbf{j} (\partial A_x / \partial z - \partial A_z / \partial x) + \mathbf{k} (\partial A_y / \partial x - \partial A_x / \partial y)$$

Let: $\mathbf{B} = \nabla \times \mathbf{A} = B_x \mathbf{i} + B_y \mathbf{j} + B_z \mathbf{k}$, where:

Component	Expression
B_x	$\partial A_z / \partial y - \partial A_y / \partial z$
B_y	$\partial A_x / \partial z - \partial A_z / \partial x$
B_z	$\partial A_y / \partial x - \partial A_x / \partial y$

3 COMPUTE THE CURL OF B — FULL DETERMINANT EXPANSION

$\nabla \times \mathbf{B} =$

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ B_x & B_y & B_z \end{vmatrix}$$

Extract the **i-component** of $(\nabla \times \mathbf{B})$:

$$\text{i-component} = \partial B_z / \partial y - \partial B_y / \partial z$$

Substituting B_z and B_y :

$$\begin{aligned} &= \partial / \partial y (\partial A_y / \partial x - \partial A_x / \partial y) - \partial / \partial z (\partial A_x / \partial z - \partial A_z / \partial x) \\ &= \partial^2 A_y / \partial y \partial x - \partial^2 A_x / \partial y^2 - \partial^2 A_x / \partial z^2 + \partial^2 A_z / \partial z \partial x \end{aligned}$$

Add and subtract $\partial^2 A_x / \partial x^2$ (a standard algebraic trick):

$$\begin{aligned} &= \partial^2 A_x / \partial x^2 + \partial^2 A_y / \partial y \partial x + \partial^2 A_z / \partial z \partial x \\ &\quad - (\partial^2 A_x / \partial x^2 + \partial^2 A_x / \partial y^2 + \partial^2 A_x / \partial z^2) \end{aligned}$$

Recognise the two grouped terms:

Group	Meaning
$\partial / \partial x (\partial A_x / \partial x + \partial A_y / \partial y + \partial A_z / \partial z)$	$= \partial / \partial x (\nabla \cdot \mathbf{A})$ – the i -component of $\nabla(\nabla \cdot \mathbf{A})$
$\partial^2 A_x / \partial x^2 + \partial^2 A_x / \partial y^2 + \partial^2 A_x / \partial z^2$	$= \nabla^2 A_x$ – the i -component of $\nabla^2 \mathbf{A}$

So the **i**-component of $(\nabla \times \mathbf{B}) = [\nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}]_x$ ✓

4 SHOW THE SAME FOR J AND K COMPONENTS (BY SYMMETRY)

By the exact same procedure applied to the **j** and **k** components (by cyclically permuting x, y, z $\rightarrow$ y, z, x $\rightarrow$ z, x, y):

Component	LHS of $\nabla \times (\nabla \times \mathbf{A})$	RHS = $\nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$ component
i	$\partial / \partial x (\nabla \cdot \mathbf{A}) - \nabla^2 A_x$	$[\nabla(\nabla \cdot \mathbf{A})]_x - [\nabla^2 \mathbf{A}]_x$
j	$\partial / \partial y (\nabla \cdot \mathbf{A}) - \nabla^2 A_y$	$[\nabla(\nabla \cdot \mathbf{A})]_y - [\nabla^2 \mathbf{A}]_y$
k	$\partial / \partial z (\nabla \cdot \mathbf{A}) - \nabla^2 A_z$	$[\nabla(\nabla \cdot \mathbf{A})]_z - [\nabla^2 \mathbf{A}]_z$

All three components of LHS match RHS.

7 FINAL CONCLUSION

✓ IDENTITY PROVED

$$\nabla \times (\nabla \times \mathbf{A}) = \nabla (\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$$

8

CONCEPT INSIGHT

Physical Importance: This is one of the most powerful identities in vector calculus. In electromagnetism, starting from Maxwell's equations, applying $\nabla \times$ to $\nabla \times \mathbf{E}$ and using this identity leads directly to the electromagnetic wave equation: $\nabla^2 \mathbf{E} - \mu_0 \epsilon_0 \partial^2 \mathbf{E} / \partial t^2 = 0$. This is how the speed of light was first derived from Maxwell's equations!

✓ IDENTITY 6 PROVED — $\nabla \times (\nabla \times \mathbf{A}) = \nabla (\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$

Laplacian of a Scalar Function

$$\nabla^2\varphi = \partial^2\varphi/\partial x^2 + \partial^2\varphi/\partial y^2 + \partial^2\varphi/\partial z^2$$

1 DEFINE VARIABLES AND GOAL

Let $\varphi(x,y,z)$ be a twice continuously differentiable scalar function. The Laplacian $\nabla^2\varphi$ is defined as the divergence of the gradient of φ :

$$\nabla^2\varphi = \nabla \cdot (\nabla\varphi)$$

We will compute this step by step using the definitions of gradient and divergence.

2 WRITE THE DEL OPERATOR AND ITS PRODUCT WITH ITSELF

The Del operator is:

$$\nabla = \mathbf{i}\partial/\partial x + \mathbf{j}\partial/\partial y + \mathbf{k}\partial/\partial z$$

The Laplacian operator $\nabla^2 = \nabla \cdot \nabla$ is the dot product of Del with itself:

$$\begin{aligned}\nabla^2 &= (\nabla \cdot \nabla) \\ &= (\mathbf{i}\partial/\partial x + \mathbf{j}\partial/\partial y + \mathbf{k}\partial/\partial z) \cdot (\mathbf{i}\partial/\partial x + \mathbf{j}\partial/\partial y + \mathbf{k}\partial/\partial z)\end{aligned}$$

Using $\mathbf{i} \cdot \mathbf{i} = 1$, $\mathbf{j} \cdot \mathbf{j} = 1$, $\mathbf{k} \cdot \mathbf{k} = 1$, and all cross products = 0:

$$\nabla^2 = \partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2$$

3 STEP 1 — COMPUTE GRADIENT OF Φ

Apply ∇ to φ :

$$\nabla\varphi = (\partial\varphi/\partial x)\mathbf{i} + (\partial\varphi/\partial y)\mathbf{j} + (\partial\varphi/\partial z)\mathbf{k}$$

This is a vector with three components, each being a partial derivative of φ .

Now apply $\nabla \cdot$ to the vector $\nabla\phi = (\partial\phi/\partial x, \partial\phi/\partial y, \partial\phi/\partial z)$:

$$\nabla \cdot (\nabla\phi) = \partial/\partial x (\partial\phi/\partial x) + \partial/\partial y (\partial\phi/\partial y) + \partial/\partial z (\partial\phi/\partial z)$$

Each term is simply a second-order partial derivative:

Term	Differentiation	Result
$\partial/\partial x$ of $(\partial\phi/\partial x)$	Differentiate $\partial\phi/\partial x$ again with respect to x	$\partial^2\phi/\partial x^2$
$\partial/\partial y$ of $(\partial\phi/\partial y)$	Differentiate $\partial\phi/\partial y$ again with respect to y	$\partial^2\phi/\partial y^2$
$\partial/\partial z$ of $(\partial\phi/\partial z)$	Differentiate $\partial\phi/\partial z$ again with respect to z	$\partial^2\phi/\partial z^2$

Adding all three terms:

$$\nabla^2\phi = \nabla \cdot (\nabla\phi) = \partial^2\phi/\partial x^2 + \partial^2\phi/\partial y^2 + \partial^2\phi/\partial z^2$$

ALTERNATIVE DERIVATION VIA OPERATOR DOT PRODUCT

Directly computing $\nabla \cdot \nabla$ acting on ϕ :

$$\begin{aligned} \nabla \cdot \nabla\phi &= (\mathbf{i}\partial/\partial x + \mathbf{j}\partial/\partial y + \mathbf{k}\partial/\partial z) \cdot (\mathbf{i}\partial\phi/\partial x + \mathbf{j}\partial\phi/\partial y + \mathbf{k}\partial\phi/\partial z) \\ &= (\mathbf{i} \cdot \mathbf{i}) \partial^2\phi/\partial x^2 + (\mathbf{i} \cdot \mathbf{j}) \partial^2\phi/\partial x\partial y + (\mathbf{i} \cdot \mathbf{k}) \partial^2\phi/\partial x\partial z \\ &\quad + (\mathbf{j} \cdot \mathbf{i}) \partial^2\phi/\partial y\partial x + (\mathbf{j} \cdot \mathbf{j}) \partial^2\phi/\partial y^2 + (\mathbf{j} \cdot \mathbf{k}) \partial^2\phi/\partial y\partial z \\ &\quad + (\mathbf{k} \cdot \mathbf{i}) \partial^2\phi/\partial z\partial x + (\mathbf{k} \cdot \mathbf{j}) \partial^2\phi/\partial z\partial y + (\mathbf{k} \cdot \mathbf{k}) \partial^2\phi/\partial z^2 \end{aligned}$$

Since $\mathbf{i} \cdot \mathbf{j} = \mathbf{i} \cdot \mathbf{k} = \mathbf{j} \cdot \mathbf{k} = 0$ and $\mathbf{i} \cdot \mathbf{i} = \mathbf{j} \cdot \mathbf{j} = \mathbf{k} \cdot \mathbf{k} = 1$, all cross terms vanish:

$$\nabla^2\phi = \partial^2\phi/\partial x^2 + \partial^2\phi/\partial y^2 + \partial^2\phi/\partial z^2 \quad \checkmark$$

FINAL CONCLUSION

✓ IDENTITY PROVED (LAPLACIAN OF A SCALAR)

$$\nabla^2\phi = \nabla \cdot (\nabla\phi) = \partial^2\phi/\partial x^2 + \partial^2\phi/\partial y^2 + \partial^2\phi/\partial z^2$$

Physical Meaning: The Laplacian $\nabla^2\phi$ measures the *net curvature* or the local average deviation of ϕ from its value at a point. It appears in many fundamental PDEs: the

Laplace equation ($\nabla^2\phi = 0$) governs electrostatics, gravitation, and steady-state heat flow; the **Poisson equation** ($\nabla^2\phi = -\rho/\epsilon_0$) governs electrostatics with sources; and the **heat equation** ($\partial\phi/\partial t = k\nabla^2\phi$) governs diffusion.

Note on Vector Laplacian: The vector Laplacian is defined as: $\nabla^2\mathbf{A} = (\nabla^2 A_x)\mathbf{i} + (\nabla^2 A_y)\mathbf{j} + (\nabla^2 A_z)\mathbf{k}$, i.e., the scalar Laplacian applied to each component. This is consistent with Identity 6 above.

✓ IDENTITY 7 PROVED — $\nabla^2\phi = \partial^2\phi/\partial x^2 + \partial^2\phi/\partial y^2 + \partial^2\phi/\partial z^2$

 Quick Reference — All Proved Vector Identities Involving ∇

MASTER FORMULA SHEET

1. $\nabla (\varphi + \psi) = \nabla \varphi + \nabla \psi$
2. $\nabla \cdot (\varphi \mathbf{A}) = \varphi (\nabla \cdot \mathbf{A}) + (\nabla \varphi) \cdot \mathbf{A}$
3. $\nabla \times (\varphi \mathbf{A}) = \varphi (\nabla \times \mathbf{A}) + (\nabla \varphi) \times \mathbf{A}$
4. $\nabla \cdot (\nabla \times \mathbf{A}) = 0$
5. $\nabla \times (\nabla \varphi) = 0$
6. $\nabla \times (\nabla \times \mathbf{A}) = \nabla (\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$
7. $\nabla^2 \varphi = \partial^2 \varphi / \partial x^2 + \partial^2 \varphi / \partial y^2 + \partial^2 \varphi / \partial z^2$

#	Identity Name	Statement	Key Tool Used in Proof	Physical Meaning
1	Gradient of Sum	$\nabla (\varphi + \psi) = \nabla \varphi + \nabla \psi$	Linearity of partial derivatives	Gradient is a linear operator (superposition)
2	Divergence of $\varphi \mathbf{A}$	$\nabla \cdot (\varphi \mathbf{A}) = \varphi (\nabla \cdot \mathbf{A}) + (\nabla \varphi) \cdot \mathbf{A}$	Product rule for partial derivatives	Mass flux divergence = compressibility + density gradient effects
3	Curl of $\varphi \mathbf{A}$	$\nabla \times (\varphi \mathbf{A}) = \varphi (\nabla \times \mathbf{A}) + (\nabla \varphi) \times \mathbf{A}$	Product rule + determinant expansion	Rotation in a scaled field decomposes into two sources
4	Div of Curl = 0	$\nabla \cdot (\nabla \times \mathbf{A}) = 0$	Clairaut's theorem (mixed partials equal)	No magnetic monopoles; curl fields are solenoidal
5	Curl of Gradient = 0	$\nabla \times (\nabla \varphi) = \mathbf{0}$	Clairaut's theorem	Conservative fields are irrotational; potential theory
6	Curl of Curl	$\nabla \times (\nabla \times \mathbf{A}) = \nabla (\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$	Full component expansion + grouping	Derives EM wave equation; core of wave physics

#	Identity Name	Statement	Key Tool Used in Proof	Physical Meaning
7	Laplacian of Scalar	$\nabla^2\phi = \partial^2\phi/\partial x^2 + \partial^2\phi/\partial y^2 + \partial^2\phi/\partial z^2$	Divergence of gradient; dot product of Del	Heat equation, Laplace/Poisson equations, wave equation



Key Theorems Used Across All Proofs

Theorem/Rule	Statement	Used In
Product Rule	$\partial (fg) / \partial x = f \partial g / \partial x + g \partial f / \partial x$	Identities 2, 3
Clairaut's Theorem	$\partial^2 f / \partial x \partial y = \partial^2 f / \partial y \partial x$ (for C^2 functions)	Identities 4, 5, 6
Linearity of $\partial/\partial x$	$\partial (f+g) / \partial x = \partial f / \partial x + \partial g / \partial x$	Identity 1
Determinant Expansion	Cofactor expansion along row 1	Identities 3, 4, 5, 6
Unit Vector Dot Products	$\mathbf{i}\cdot\mathbf{i}=\mathbf{j}\cdot\mathbf{j}=\mathbf{k}\cdot\mathbf{k}=1$; cross-terms = 0	Identity 7

End of Section — All 7 Vector Identities Involving ∇ Proved

Every step derived from first principles • No shortcuts • Exam-Ready Format

Section 14 – Applications of Vector Differentiation

Application 1: Finding Normal to a Surface

If a surface is given by $\varphi(x,y,z) = c$, the unit normal at any point is $\hat{n} = \nabla\varphi/|\nabla\varphi|$.

This is used in computer graphics (shading), fluid mechanics (boundary conditions), and structural engineering (stress directions).

Application 2: Finding Angle Between Surfaces

Two surfaces $\varphi_1 = c_1$ and $\varphi_2 = c_2$ intersect along a curve. The angle θ between the surfaces satisfies:

$$\cos \theta = (\nabla\varphi_1 \cdot \nabla\varphi_2) / (|\nabla\varphi_1| \times |\nabla\varphi_2|)$$

Application 3: Gauss's Divergence Theorem

$$\iiint_V \nabla \cdot \mathbf{F} \, dV = \iint_S \mathbf{F} \cdot \hat{n} \, dS$$

The surface integral of a vector field over a closed surface equals the volume integral of its divergence. Used in fluid dynamics and electrostatics.

Application 4: Stokes' Theorem

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \hat{n} \, dS$$

The line integral of $\mathbf{F}$ around a closed curve equals the surface integral of curl $\mathbf{F}$ over any surface bounded by that curve.

Application 5: Irrotational and Solenoidal Fields

Condition	Name	Physical Example
$\nabla \cdot \mathbf{F} = 0$	Solenoidal field	Incompressible fluid flow, magnetic field ($\nabla \cdot \mathbf{B} = 0$)
$\nabla \times \mathbf{F} = 0$	Irrotational field	Electrostatic field, gravitational field
Both = 0	Harmonic field	Laplace equation solutions

Example 1 – Find the Gradient of $\phi = x^2 + y^2 + z^2$

STEP 1: GIVEN FUNCTION

We are given the scalar function: $\phi(x, y, z) = x^2 + y^2 + z^2$

Physically, this is the square of the distance from the origin to any point (x, y, z) in space.

STEP 2: IDENTIFY THE OPERATION

We need to find $\nabla\phi$ — the **gradient** of ϕ . The gradient of a scalar function gives us a vector field showing the direction and rate of maximum increase.

STEP 3: WRITE THE FORMULA

$$\nabla\phi = (\partial\phi/\partial x)\mathbf{i} + (\partial\phi/\partial y)\mathbf{j} + (\partial\phi/\partial z)\mathbf{k}$$

We need to compute three partial derivatives — one for each variable.

STEP 4: EXPAND THE DEL OPERATOR

$$\nabla = \mathbf{i}(\partial/\partial x) + \mathbf{j}(\partial/\partial y) + \mathbf{k}(\partial/\partial z)$$

This means we will take partial derivative with respect to x (treating y, z as constant), then y (treating x, z as constant), then z (treating x, y as constant).

STEP 5: COMPUTE PARTIAL DERIVATIVES ONE BY ONE

$$\partial\phi/\partial x: \phi = x^2 + y^2 + z^2$$

Treating y and z as constants: $d(x^2)/dx = 2x$; $d(y^2)/dx = 0$ (constant); $d(z^2)/dx = 0$ (constant)

$$\rightarrow \partial\phi/\partial x = 2x$$

$$\partial\phi/\partial y: \phi = x^2 + y^2 + z^2$$

Treating x and z as constants: $d(x^2)/dy = 0$; $d(y^2)/dy = 2y$; $d(z^2)/dy = 0$

$$\rightarrow \partial\phi/\partial y = 2y$$

$$\partial\phi/\partial z: \phi = x^2 + y^2 + z^2$$

Treating x and y as constants: $d(x^2)/dz = 0$; $d(y^2)/dz = 0$; $d(z^2)/dz = 2z$

$$\rightarrow \partial\phi/\partial z = 2z$$

STEP 6: SUBSTITUTE AND ASSEMBLE

$$\nabla\phi = (2x)\mathbf{i} + (2y)\mathbf{j} + (2z)\mathbf{k}$$

STEP 7: SIMPLIFY

$$\nabla\phi = 2(x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) = 2\mathbf{r}$$

where $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ is the position vector.

STEP 9: PHYSICAL INTERPRETATION

The gradient of $x^2 + y^2 + z^2$ points radially outward from the origin, in the direction of $\mathbf{r}$. This makes sense because $x^2 + y^2 + z^2$ increases most rapidly as you move directly away from the origin.

STEP 10: FINAL ANSWER

$$\nabla\phi = 2xi + 2yj + 2zk = 2\mathbf{r}$$

STEP 11: CONCEPT SUMMARY

$\nabla(r^2) = 2\mathbf{r}$. In general, $\nabla(r^n) = nr^{n-2}\mathbf{r}$. The gradient of any radially symmetric function points in the radial direction.

Example 2 – Find $\nabla\phi$ for $\phi = x^2y + y^2z + z^2x$ at point $P(1, 1, 1)$

STEP 1: GIVEN FUNCTION

$\phi(x, y, z) = x^2y + y^2z + z^2x$. This is a polynomial scalar function of three variables.

STEP 2: IDENTIFY OPERATION

We find $\nabla\phi$ (gradient) and then evaluate at $P(1,1,1)$.

STEP 3: FORMULA

$$\nabla\phi = (\partial\phi/\partial x)\mathbf{i} + (\partial\phi/\partial y)\mathbf{j} + (\partial\phi/\partial z)\mathbf{k}$$

STEP 5: COMPUTE PARTIAL DERIVATIVES

$\partial\phi/\partial x$: $\phi = x^2y + y^2z + z^2x$

- $d(x^2y)/dx = 2xy$ (y is constant, derivative of x^2 is $2x$)
- $d(y^2z)/dx = 0$ (no x term, treating as constant)
- $d(z^2x)/dx = z^2$ (z is constant, derivative of x is 1)

$$\rightarrow \partial\phi/\partial x = 2xy + z^2$$

$\partial\phi/\partial y$: $\phi = x^2y + y^2z + z^2x$

- $d(x^2y)/dy = x^2$ (x is constant, derivative of y is 1)
- $d(y^2z)/dy = 2yz$ (z is constant, derivative of y^2 is $2y$)
- $d(z^2x)/dy = 0$ (no y term)

$$\rightarrow \partial\phi/\partial y = x^2 + 2yz$$

$\partial\phi/\partial z$: $\phi = x^2y + y^2z + z^2x$

- $d(x^2y)/dz = 0$ (no z term)
- $d(y^2z)/dz = y^2$ (y is constant, derivative of z is 1)
- $d(z^2x)/dz = 2zx$ (x is constant, derivative of z^2 is $2z$)

$$\rightarrow \partial\phi/\partial z = y^2 + 2zx$$

STEP 6: ASSEMBLE GRADIENT

$$\nabla\phi = (2xy + z^2)\mathbf{i} + (x^2 + 2yz)\mathbf{j} + (y^2 + 2zx)\mathbf{k}$$

STEP 7: EVALUATE AT $P(1, 1, 1)$ – SUBSTITUTE $X=1, Y=1, Z=1$

i-component: $2(1)(1) + (1)^2 = 2 + 1 = 3$

j-component: $(1)^2 + 2(1)(1) = 1 + 2 = 3$

k-component: $(1)^2 + 2(1)(1) = 1 + 2 = 3$

STEP 10: FINAL ANSWER

$$\nabla\phi \text{ at } P(1,1,1) = 3\mathbf{i} + 3\mathbf{j} + 3\mathbf{k}$$

The magnitude: $|\nabla\varphi| = \sqrt{9+9+9} = \sqrt{27} = 3\sqrt{3}$

Example 3 – Directional Derivative of $\phi = x^2yz + 4xz^2$ at $(1, -2, -1)$ in direction of $2\mathbf{i} - \mathbf{j} - 2\mathbf{k}$

STEP 1: GIVEN

$\phi = x^2yz + 4xz^2$, Point $P = (1, -2, -1)$, Direction vector: $\mathbf{a} = 2\mathbf{i} - \mathbf{j} - 2\mathbf{k}$

STEP 2: FORMULA FOR DIRECTIONAL DERIVATIVE

$D_{\mathbf{a}} \phi = \nabla \phi \cdot \hat{\mathbf{a}}$ where $\hat{\mathbf{a}} = \mathbf{a}/|\mathbf{a}|$ is the unit vector in the direction of $\mathbf{a}$.

STEP 5: COMPUTE PARTIAL DERIVATIVES OF Φ

$\partial \phi / \partial x$: $\phi = x^2yz + 4xz^2$

- $d(x^2yz)/dx = 2xyz$ (yz is constant, derivative of x^2 is $2x$)
- $d(4xz^2)/dx = 4z^2$ ($4z^2$ is constant coefficient, derivative of x is 1)

$$\rightarrow \partial \phi / \partial x = 2xyz + 4z^2$$

$\partial \phi / \partial y$: $\phi = x^2yz + 4xz^2$

- $d(x^2yz)/dy = x^2z$ (x^2z is constant, derivative of y is 1)
- $d(4xz^2)/dy = 0$ (no y present)

$$\rightarrow \partial \phi / \partial y = x^2z$$

$\partial \phi / \partial z$: $\phi = x^2yz + 4xz^2$

- $d(x^2yz)/dz = x^2y$ (x^2y is constant, derivative of z is 1)
- $d(4xz^2)/dz = 8xz$ ($4x$ is constant, derivative of z^2 is $2z \rightarrow 4x \cdot 2z = 8xz$)

$$\rightarrow \partial \phi / \partial z = x^2y + 8xz$$

STEP 6: EVALUATE $\nabla \Phi$ AT $P(1, -2, -1)$

$$\partial \phi / \partial x \text{ at } (1, -2, -1) = 2(1)(-2)(-1) + 4(-1)^2 = 4 + 4 = 8$$

$$\partial \phi / \partial y \text{ at } (1, -2, -1) = (1)^2(-1) = -1$$

$$\partial \phi / \partial z \text{ at } (1, -2, -1) = (1)^2(-2) + 8(1)(-1) = -2 - 8 = -10$$

$$\text{So } \nabla \phi \text{ at } P = 8\mathbf{i} - \mathbf{j} - 10\mathbf{k}$$

STEP 7: FIND UNIT VECTOR $\hat{\mathbf{A}}$

$$\mathbf{a} = 2\mathbf{i} - \mathbf{j} - 2\mathbf{k}$$

$$|\mathbf{a}| = \sqrt{(2)^2 + (-1)^2 + (-2)^2} = \sqrt{4 + 1 + 4} = \sqrt{9} = 3$$

$$\hat{\mathbf{a}} = (2\mathbf{i} - \mathbf{j} - 2\mathbf{k})/3 = (2/3)\mathbf{i} - (1/3)\mathbf{j} - (2/3)\mathbf{k}$$

STEP 8: COMPUTE DOT PRODUCT (DIRECTIONAL DERIVATIVE)

$$D_{\mathbf{a}} \phi = \nabla \phi \cdot \hat{\mathbf{a}} = (8\mathbf{i} - \mathbf{j} - 10\mathbf{k}) \cdot ((2/3)\mathbf{i} - (1/3)\mathbf{j} - (2/3)\mathbf{k})$$

$$= 8 \times (2/3) + (-1) \times (-1/3) + (-10) \times (-2/3)$$

$$= 16/3 + 1/3 + 20/3$$

$$= (16 + 1 + 20)/3$$

$$= 37/3$$

STEP 10: FINAL ANSWER

$$\text{Directional Derivative} = 37/3 \approx 12.33$$

Example 4 – Find the Unit Normal to the Surface $x^2y + 2xz = 4$ at $(2, -2, 3)$

STEP 1: IDENTIFY THE SURFACE

$\phi(x,y,z) = x^2y + 2xz$. The surface is defined by $\phi = 4$.

STEP 5: COMPUTE PARTIAL DERIVATIVES

$$\partial\phi/\partial x = 2xy + 2z$$

$$\partial\phi/\partial y = x^2$$

$$\partial\phi/\partial z = 2x$$

STEP 6: EVALUATE AT $(2, -2, 3)$

$$\partial\phi/\partial x = 2(2)(-2) + 2(3) = -8 + 6 = -2$$

$$\partial\phi/\partial y = (2)^2 = 4$$

$$\partial\phi/\partial z = 2(2) = 4$$

$$\nabla\phi = -2\mathbf{i} + 4\mathbf{j} + 4\mathbf{k}$$

STEP 7: MAGNITUDE OF $\nabla\phi$

$$|\nabla\phi| = \sqrt{(-2)^2 + 4^2 + 4^2} = \sqrt{4 + 16 + 16} = \sqrt{36} = 6$$

STEP 8: UNIT NORMAL VECTOR

$$\hat{n} = \nabla\phi / |\nabla\phi| = (-2\mathbf{i} + 4\mathbf{j} + 4\mathbf{k}) / 6$$

$$\hat{n} = (-1/3)\mathbf{i} + (2/3)\mathbf{j} + (2/3)\mathbf{k}$$

STEP 10: FINAL ANSWER

$$\text{Unit Normal } \hat{n} = (-\mathbf{i} + 2\mathbf{j} + 2\mathbf{k})/3$$

Example 5 – Find the Divergence of $\mathbf{F} = x^2\mathbf{i} + y^2\mathbf{j} + z^2\mathbf{k}$

STEP 1: GIVEN

$\mathbf{F} = x^2\mathbf{i} + y^2\mathbf{j} + z^2\mathbf{k}$. Components: $F_1 = x^2$, $F_2 = y^2$, $F_3 = z^2$

STEP 2: IDENTIFY OPERATION

We need divergence: $\nabla \cdot \mathbf{F} = \partial F_1 / \partial x + \partial F_2 / \partial y + \partial F_3 / \partial z$

Result will be a scalar.

STEP 5: COMPUTE EACH TERM

$\partial F_1 / \partial x$: $F_1 = x^2$. Differentiate w.r.t. x : $\rightarrow 2x$

$\partial F_2 / \partial y$: $F_2 = y^2$. Differentiate w.r.t. y : $\rightarrow 2y$

$\partial F_3 / \partial z$: $F_3 = z^2$. Differentiate w.r.t. z : $\rightarrow 2z$

STEP 6: ADD ALL THREE TERMS

$$\nabla \cdot \mathbf{F} = 2x + 2y + 2z = 2(x + y + z)$$

STEP 9: INTERPRETATION

The divergence is $2(x+y+z)$. This is positive when $x+y+z > 0$, meaning fluid is flowing outward (source) in that region. It is zero on the plane $x+y+z = 0$.

STEP 10: FINAL ANSWER

$$\nabla \cdot \mathbf{F} = 2(\mathbf{x} + \mathbf{y} + \mathbf{z})$$

Example 6 – Find $\nabla \cdot \mathbf{F}$ where $\mathbf{F} = x^2y\mathbf{i} + xy^2z\mathbf{j} + z^3\mathbf{k}$

STEP 1: GIVEN

$$\mathbf{F} = x^2y\mathbf{i} + xy^2z\mathbf{j} + z^3\mathbf{k}$$

$$F_1 = x^2y, F_2 = xy^2z, F_3 = z^3$$

STEP 5: PARTIAL DERIVATIVES

$$\partial F_1 / \partial x: F_1 = x^2y. \text{ Treat } y \text{ as constant. } d(x^2)/dx = 2x. \text{ So: } 2xy$$

$$\partial F_2 / \partial y: F_2 = xy^2z. \text{ Treat } x, z \text{ as constant. } d(y^2)/dy = 2y. \text{ So: } 2xyz$$

$$\partial F_3 / \partial z: F_3 = z^3. \text{ Treat } x, y \text{ as constant. } d(z^3)/dz = 3z^2. \text{ So: } 3z^2$$

STEP 6: SUM

$$\nabla \cdot \mathbf{F} = 2xy + 2xyz + 3z^2$$

STEP 10: FINAL ANSWER

$$\nabla \cdot \mathbf{F} = 2xy + 2xyz + 3z^2$$

Example 7 – Show that $\mathbf{F} = (y^2 - z^2 + 3yz - 2x)\mathbf{i} + (3xz + 2xy)\mathbf{j} + (3xy - 2xz + 2z)\mathbf{k}$ is Solenoidal

STEP 1: GIVEN

$$\mathbf{F} = (y^2 - z^2 + 3yz - 2x)\mathbf{i} + (3xz + 2xy)\mathbf{j} + (3xy - 2xz + 2z)\mathbf{k}$$

$$F_1 = y^2 - z^2 + 3yz - 2x, F_2 = 3xz + 2xy, F_3 = 3xy - 2xz + 2z$$

STEP 2: FOR SOLENOIDAL FIELD

A field is solenoidal if $\nabla \cdot \mathbf{F} = 0$. We must prove $\partial F_1 / \partial x + \partial F_2 / \partial y + \partial F_3 / \partial z = 0$.

STEP 5: COMPUTE PARTIAL DERIVATIVES

$$\partial \mathbf{F}_1 / \partial \mathbf{x}: F_1 = y^2 - z^2 + 3yz - 2x$$

Only $-2x$ depends on x . $\rightarrow \partial F_1 / \partial x = -2$

$$\partial \mathbf{F}_2 / \partial \mathbf{y}: F_2 = 3xz + 2xy$$

Only $2xy$ depends on y . $\rightarrow \partial F_2 / \partial y = 2x$

$$\partial \mathbf{F}_3 / \partial \mathbf{z}: F_3 = 3xy - 2xz + 2z$$

$$-2xz \rightarrow -2x; 2z \rightarrow 2 \rightarrow \partial F_3 / \partial z = -2x + 2$$

STEP 6: SUM ALL THREE

$$\nabla \cdot \mathbf{F} = (-2) + (2x) + (-2x + 2)$$

$$= -2 + 2x - 2x + 2$$

$$= 0$$

STEP 10: FINAL ANSWER

$\nabla \cdot \mathbf{F} = 0$, therefore $\mathbf{F}$ is a **Solenoidal** vector field. ✓

Example 8 – Find Curl of $F = (x^2 - y^2)\mathbf{i} + 2xy\mathbf{j} + 0\mathbf{k}$

STEP 1: GIVEN

$$F = (x^2 - y^2)\mathbf{i} + 2xy\mathbf{j} + 0\mathbf{k}$$

$$F_1 = x^2 - y^2, F_2 = 2xy, F_3 = 0$$

STEP 2: SET UP THE DETERMINANT

$$\nabla \times F = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2 - y^2 & 2xy & 0 \end{vmatrix}$$

STEP 5: EXPAND THE DETERMINANT (ROW 1, COFACTORS)

i-component (cofactor M_{11}):

Minor: $|\partial/\partial y \ \partial/\partial z|$ applied to $|2xy \ 0|$

$$= \partial(0)/\partial y - \partial(2xy)/\partial z = 0 - 0 = 0$$

j-component (cofactor M_{12} , note the NEGATIVE sign):

Minor: $|\partial/\partial x \ \partial/\partial z|$ applied to $|x^2 - y^2 \ 0|$

$$= -[\partial(0)/\partial x - \partial(x^2 - y^2)/\partial z] = -[0 - 0] = 0$$

k-component (cofactor M_{13}):

Minor: $|\partial/\partial x \ \partial/\partial y|$ applied to $|x^2 - y^2 \ 2xy|$

$$= \partial(2xy)/\partial x - \partial(x^2 - y^2)/\partial y$$

$$= 2y - (-2y) = 2y + 2y = 4y$$

STEP 6: ASSEMBLE CURL

$$\nabla \times F = 0\mathbf{i} + 0\mathbf{j} + 4y\mathbf{k} = 4y\mathbf{k}$$

STEP 9: INTERPRETATION

The curl is in the k-direction (vertical axis). This means the vector field has a rotational component around the z-axis. The magnitude of rotation is proportional to y.

STEP 10: FINAL ANSWER

$$\nabla \times F = 4y\mathbf{k}$$

Example 9 – Find Curl of $F = (x+2y+az)\mathbf{i} + (bx-3y-z)\mathbf{j} + (4x+cy+2z)\mathbf{k}$. Find a,b,c if F is irrotational.

STEP 1: GIVEN

$$F_1 = x+2y+az, F_2 = bx-3y-z, F_3 = 4x+cy+2z$$

$$\text{For irrotational: } \nabla \times F = 0$$

STEP 5: COMPUTE CURL COMPONENTS

$$\textbf{i-component: } \partial F_3 / \partial y - \partial F_2 / \partial z = c - (-1) = c + 1$$

$$\textbf{j-component (with - sign): } -(\partial F_3 / \partial x - \partial F_1 / \partial z) = -(4 - a) = a - 4$$

$$\textbf{k-component: } \partial F_2 / \partial x - \partial F_1 / \partial y = b - 2$$

STEP 6: SET EACH COMPONENT = 0

For $\nabla \times F = 0$, each component must be zero:

$$c + 1 = 0 \rightarrow \mathbf{c = -1}$$

$$a - 4 = 0 \rightarrow \mathbf{a = 4}$$

$$b - 2 = 0 \rightarrow \mathbf{b = 2}$$

STEP 10: FINAL ANSWER

$$\mathbf{a = 4, b = 2, c = -1}$$

Example 10 – Full Curl Expansion: $\mathbf{F} = yz\mathbf{i} + zx\mathbf{j} + xy\mathbf{k}$

STEP 1: GIVEN

$$\mathbf{F} = yz\mathbf{i} + zx\mathbf{j} + xy\mathbf{k}. \quad F_1 = yz, \quad F_2 = zx, \quad F_3 = xy$$

STEP 2: SET UP DETERMINANT

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ yz & zx & xy \end{vmatrix}$$

STEP 5: EXPAND EACH MINOR CAREFULLY

i-component: $\partial(xy)/\partial y - \partial(zx)/\partial z = x - x = 0$

j-component (– sign!): $-\partial(xy)/\partial x - \partial(yz)/\partial z = -[y - y] = 0$

k-component: $\partial(zx)/\partial x - \partial(yz)/\partial y = z - z = 0$

STEP 10: FINAL ANSWER

$\nabla \times \mathbf{F} = \mathbf{0}$. The field $\mathbf{F} = yz\mathbf{i} + zx\mathbf{j} + xy\mathbf{k}$ is **irrotational!**

STEP 11: NOTE

This also means $\mathbf{F}$ can be written as $\nabla\phi$ for some scalar ϕ . In fact, $\phi = xyz$ (you can verify $\nabla(xyz) = yz\mathbf{i} + xz\mathbf{j} + xy\mathbf{k} = \mathbf{F}$).

Example 11 – Verify $\nabla \times (\nabla\phi) = \mathbf{0}$ for $\phi = x^2y + y^2z$

STEP 1: FIND $\nabla\phi$ FIRST

$$\phi = x^2y + y^2z$$

$$\partial\phi/\partial x = 2xy, \quad \partial\phi/\partial y = x^2 + 2yz, \quad \partial\phi/\partial z = y^2$$

$$\nabla\phi = 2xy\mathbf{i} + (x^2 + 2yz)\mathbf{j} + y^2\mathbf{k}$$

Let $\mathbf{G} = \nabla\phi$. So $G_1 = 2xy$, $G_2 = x^2 + 2yz$, $G_3 = y^2$

STEP 5: COMPUTE $\nabla \times \mathbf{G}$

i-component: $\partial G_3/\partial y - \partial G_2/\partial z = \partial(y^2)/\partial y - \partial(x^2 + 2yz)/\partial z = 2y - 2y = 0$

j-component (– sign!): $-\partial G_3/\partial x - \partial G_1/\partial z = -[\partial(y^2)/\partial x - \partial(2xy)/\partial z] = -[0 - 0] = 0$

k-component: $\partial G_2/\partial x - \partial G_1/\partial y = \partial(x^2 + 2yz)/\partial x - \partial(2xy)/\partial y = 2x - 2x = 0$

STEP 10: FINAL ANSWER

$\nabla \times (\nabla\phi) = 0\mathbf{i} + 0\mathbf{j} + 0\mathbf{k} = \mathbf{0}$ ✓ Identity verified!

Example 12 – Find Max Directional Derivative of $\phi = xyz^2$ at $(1, 0, 3)$

STEP 5: COMPUTE PARTIAL DERIVATIVES

$$\partial\phi/\partial x = yz^2, \partial\phi/\partial y = xz^2, \partial\phi/\partial z = 2xyz$$

STEP 6: EVALUATE AT $(1, 0, 3)$

$$\partial\phi/\partial x = (0)(9) = 0$$

$$\partial\phi/\partial y = (1)(9) = 9$$

$$\partial\phi/\partial z = 2(1)(0)(3) = 0$$

$$\nabla\phi = 0\mathbf{i} + 9\mathbf{j} + 0\mathbf{k} = 9\mathbf{j}$$

STEP 7: MAXIMUM DIRECTIONAL DERIVATIVE = $|\nabla\phi|$

$$|\nabla\phi| = |9\mathbf{j}| = 9$$

STEP 10: FINAL ANSWER

Maximum Directional Derivative = 9, in the direction $\mathbf{j}$ (y-axis)

Example 13 – Find Divergence and Curl of $\mathbf{F} = (x^3 + y^3)\mathbf{i} + 3xy^2\mathbf{j} - 6x^2y\mathbf{k}$

STEP 1: COMPONENTS

$$F_1 = x^3 + y^3, F_2 = 3xy^2, F_3 = -6x^2y$$

STEP 5A: DIVERGENCE

$$\partial F_1/\partial x = 3x^2, \partial F_2/\partial y = 6xy, \partial F_3/\partial z = 0$$

$$\nabla \cdot \mathbf{F} = 3x^2 + 6xy$$

STEP 5B: CURL

$$\mathbf{i}\text{-component: } \partial F_3/\partial y - \partial F_2/\partial z = \partial(-6x^2y)/\partial y - \partial(3xy^2)/\partial z = -6x^2 - 0 = -6x^2$$

$$\mathbf{j}\text{-component (– sign): } -[\partial F_3/\partial x - \partial F_1/\partial z] = -[\partial(-6x^2y)/\partial x - 0] = -[-12xy] = 12xy$$

$$\mathbf{k}\text{-component: } \partial F_2/\partial x - \partial F_1/\partial y = \partial(3xy^2)/\partial x - \partial(x^3 + y^3)/\partial y = 3y^2 - 3y^2 = 0$$

STEP 10: FINAL ANSWER

$$\nabla \cdot \mathbf{F} = 3x^2 + 6xy$$

$$\nabla \times \mathbf{F} = -6x^2\mathbf{i} + 12xy\mathbf{j} + 0\mathbf{k}$$

Example 14 – Find angle between surfaces $x^2+y^2+z^2=9$ and $x^2+y^2-z=3$ at $(2,-1,2)$

STEP 1

$$\varphi_1 = x^2+y^2+z^2-9, \varphi_2 = x^2+y^2-z-3$$

STEP 5: GRADIENTS

$$\nabla\varphi_1 = 2xi+2yj+2zk \rightarrow \text{at } (2,-1,2): 4i-2j+4k$$

$$\nabla\varphi_2 = 2xi+2yj-k \rightarrow \text{at } (2,-1,2): 4i-2j-k$$

STEP 7: ANGLE

$$\cos \theta = (\nabla\varphi_1 \cdot \nabla\varphi_2)/(|\nabla\varphi_1||\nabla\varphi_2|)$$

$$\nabla\varphi_1 \cdot \nabla\varphi_2 = 16+4-4 = 16$$

$$|\nabla\varphi_1| = \sqrt{(16+4+16)} = 6, |\nabla\varphi_2| = \sqrt{(16+4+1)} = \sqrt{21}$$

$$\cos \theta = 16/(6\sqrt{21}) = 8/(3\sqrt{21})$$

$$\theta = \cos^{-1}(8/3\sqrt{21})$$

FINAL ANSWER

$$\theta = \cos^{-1}(8/3\sqrt{21})$$

Example 15 – Show $F = (2xy+z^3)i + x^2j + 3xz^2k$ is irrotational; find scalar potential

STEP 1

$$F_1=2xy+z^3, F_2=x^2, F_3=3xz^2$$

STEP 5: COMPUTE CURL

$$i: \partial F_3/\partial y - \partial F_2/\partial z = 0 - 0 = 0$$

$$j: -(\partial F_3/\partial x - \partial F_1/\partial z) = -(3z^2 - 3z^2) = 0$$

$$k: \partial F_2/\partial x - \partial F_1/\partial y = 2x - 2x = 0$$

$$\nabla \times F = 0 \rightarrow F \text{ is irrotational } \checkmark$$

STEP 6: FIND SCALAR POTENTIAL Φ SUCH THAT $F = \nabla\Phi$

$$\partial\varphi/\partial x = 2xy+z^3 \rightarrow \varphi = x^2y + xz^3 + f(y,z)$$

$$\partial\varphi/\partial y = x^2 \rightarrow x^2 + \partial f/\partial y = x^2 \rightarrow \partial f/\partial y = 0 \rightarrow f = g(z)$$

$$\partial\varphi/\partial z = 3xz^2 \rightarrow 3xz^2 + g'(z) = 3xz^2 \rightarrow g'(z) = 0 \rightarrow g = C$$

FINAL ANSWER

$$\varphi = x^2y + xz^3 + C$$

Example 16 – Verify $\nabla \cdot (\nabla \times \mathbf{F}) = 0$ for $\mathbf{F} = x^2y\mathbf{i} + y^2z\mathbf{j} + z^2x\mathbf{k}$

STEP 1

$$\mathbf{F}_1 = x^2y, \mathbf{F}_2 = y^2z, \mathbf{F}_3 = z^2x$$

STEP 5: COMPUTE CURL

$$\text{i: } \partial(z^2x)/\partial y - \partial(y^2z)/\partial z = 0 - y^2 = -y^2$$

$$\text{j: } -[\partial(z^2x)/\partial x - \partial(x^2y)/\partial z] = -[z^2 - 0] = -z^2$$

$$\text{k: } \partial(y^2z)/\partial x - \partial(x^2y)/\partial y = 0 - x^2 = -x^2$$

$$\nabla \times \mathbf{F} = -y^2\mathbf{i} - z^2\mathbf{j} - x^2\mathbf{k}$$

STEP 6: DIVERGENCE OF $(\nabla \times \mathbf{F})$

$$\text{Let } \mathbf{G} = -y^2\mathbf{i} - z^2\mathbf{j} - x^2\mathbf{k}$$

$$\partial G_1/\partial x = \partial(-y^2)/\partial x = 0$$

$$\partial G_2/\partial y = \partial(-z^2)/\partial y = 0$$

$$\partial G_3/\partial z = \partial(-x^2)/\partial z = 0$$

$$\nabla \cdot \mathbf{G} = 0 + 0 + 0 = 0 \quad \checkmark$$

FINAL ANSWER

$$\nabla \cdot (\nabla \times \mathbf{F}) = 0 \quad (\text{Identity verified!})$$

Example 17 – Gradient of $\varphi = \log(x^2 + y^2 + z^2)$

STEP 1

$$\varphi = \log(x^2 + y^2 + z^2) = \log(r^2) \text{ where } r^2 = x^2 + y^2 + z^2$$

STEP 5: PARTIAL DERIVATIVES (USING CHAIN RULE)

$$\partial \varphi / \partial x = (1/(x^2 + y^2 + z^2)) \cdot 2x = 2x/r^2$$

$$\partial \varphi / \partial y = 2y/r^2$$

$$\partial \varphi / \partial z = 2z/r^2$$

FINAL ANSWER

$$\nabla \varphi = (2/r^2)(x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) = 2\mathbf{r}/r^2 = 2\hat{\mathbf{r}}/r$$

Example 18 – Prove $\nabla(\phi/\psi) = (\psi\nabla\phi - \phi\nabla\psi)/\psi^2$

STEP 2: APPLY QUOTIENT RULE

Consider the x-component of $\nabla(\phi/\psi)$:

$$\partial(\phi/\psi)/\partial x = [\psi(\partial\phi/\partial x) - \phi(\partial\psi/\partial x)] / \psi^2$$

Similarly for y and z components.

STEP 6: ASSEMBLE VECTOR

$$\begin{aligned}\nabla(\phi/\psi) &= (1/\psi^2)[\psi(\partial\phi/\partial x \mathbf{i} + \partial\phi/\partial y \mathbf{j} + \partial\phi/\partial z \mathbf{k}) - \phi(\partial\psi/\partial x \mathbf{i} + \partial\psi/\partial y \mathbf{j} + \partial\psi/\partial z \mathbf{k})] \\ &= (\psi\nabla\phi - \phi\nabla\psi)/\psi^2 \quad \checkmark\end{aligned}$$

FINAL ANSWER

$$\nabla(\phi/\psi) = (\psi\nabla\phi - \phi\nabla\psi)/\psi^2$$

Example 19 – Divergence of $\mathbf{F} = \mathbf{r}/r^3$ (inverse square law field)

STEP 1

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}, \quad r = |\mathbf{r}| = \sqrt{(x^2 + y^2 + z^2)}, \quad \mathbf{F} = (x/r^3)\mathbf{i} + (y/r^3)\mathbf{j} + (z/r^3)\mathbf{k}$$

STEP 5: COMPUTE $\partial F_1/\partial x$ USING QUOTIENT RULE

$$F_1 = x \cdot r^{-3} = x(x^2 + y^2 + z^2)^{-3/2}$$

$$\partial F_1/\partial x = r^{-3} + x \cdot (-3/2) \cdot 2x \cdot r^{-5} = r^{-3} - 3x^2 r^{-5}$$

$$\text{Similarly: } \partial F_2/\partial y = r^{-3} - 3y^2 r^{-5}, \quad \partial F_3/\partial z = r^{-3} - 3z^2 r^{-5}$$

STEP 6: SUM

$$\nabla \cdot \mathbf{F} = 3r^{-3} - 3(x^2 + y^2 + z^2)r^{-5} = 3r^{-3} - 3r^2 \cdot r^{-5} = 3r^{-3} - 3r^{-3} = 0$$

FINAL ANSWER

$\nabla \cdot (\mathbf{r}/r^3) = 0$ for $\mathbf{r} \neq \mathbf{0}$. This represents an incompressible, purely radial flow (like gravity or electric field).

Example 20 – Full Curl: $\mathbf{F} = (3x^2+6y)\mathbf{i} - 14yz\mathbf{j} + 20xz^2\mathbf{k}$

STEP 1

$$F_1=3x^2+6y, F_2=-14yz, F_3=20xz^2$$

STEP 2: DETERMINANT SETUP

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 3x^2+6y & -14yz & 20xz^2 \end{vmatrix}$$

STEP 5: EACH COMPONENT

$$\mathbf{i}: \partial(20xz^2)/\partial y - \partial(-14yz)/\partial z = 0 - (-14) = 14$$

$$\mathbf{j} \text{ (with } -): -[\partial(20xz^2)/\partial x - \partial(3x^2+6y)/\partial z] = -[20z^2 - 0] = -20z^2$$

$$\mathbf{k}: \partial(-14yz)/\partial x - \partial(3x^2+6y)/\partial y = 0 - 6 = -6$$

FINAL ANSWER

$$\nabla \times \mathbf{F} = 14y\mathbf{i} - 20z^2\mathbf{j} - 6\mathbf{k}$$

Example 21 – Gradient of $\varphi = e^{(x^2+y^2+z^2)}$

STEP 5: CHAIN RULE DERIVATIVES

$$\text{Let } u = x^2+y^2+z^2, \text{ so } \varphi = e^u$$

$$\partial\varphi/\partial x = e^u \cdot 2x = 2x \cdot e^{(x^2+y^2+z^2)}$$

$$\partial\varphi/\partial y = 2y \cdot e^{(x^2+y^2+z^2)}$$

$$\partial\varphi/\partial z = 2z \cdot e^{(x^2+y^2+z^2)}$$

FINAL ANSWER

$$\nabla\varphi = 2(x\mathbf{i}+y\mathbf{j}+z\mathbf{k}) \cdot e^{(\mathbf{r}^2)} = 2\mathbf{r} \cdot e^{(\mathbf{r}^2)}$$

Example 22 – Prove $\nabla(r^n) = nr^{n-2}\mathbf{r}$ where $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$

STEP 5: PARTIAL DERIVATIVE

$$r^n = (x^2 + y^2 + z^2)^{n/2}$$

$$\partial(r^n)/\partial x = (n/2)(x^2 + y^2 + z^2)^{n/2-1} \cdot 2x = n \cdot x \cdot r^{n-2}$$

$$\text{Similarly: } \partial(r^n)/\partial y = n \cdot y \cdot r^{n-2}, \partial(r^n)/\partial z = n \cdot z \cdot r^{n-2}$$

STEP 6: ASSEMBLE

$$\nabla(r^n) = nr^{n-2}(x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) = nr^{n-2}\mathbf{r} \quad \checkmark$$

FINAL ANSWER

$$\nabla(r^n) = nr^{n-2}\mathbf{r}$$

Example 23 – Verify vector identity $\nabla \cdot (\phi \mathbf{F}) = \phi(\nabla \cdot \mathbf{F}) + (\nabla \phi) \cdot \mathbf{F}$

STEP 1: LET

$$\phi = x^2, \mathbf{F} = y\mathbf{i} + z\mathbf{j} + x\mathbf{k}, F_1=y, F_2=z, F_3=x$$

STEP 5: LHS – COMPUTE $\phi \mathbf{F}$ THEN DIVERGENCE

$$\phi \mathbf{F} = x^2 y \mathbf{i} + x^2 z \mathbf{j} + x^3 \mathbf{k}$$

$$\nabla \cdot (\phi \mathbf{F}) = \partial(x^2 y)/\partial x + \partial(x^2 z)/\partial y + \partial(x^3)/\partial z = 2xy + 0 + 0 = 2xy$$

STEP 6: RHS – TWO SEPARATE PARTS

$$\phi(\nabla \cdot \mathbf{F}) = x^2(\partial y/\partial x + \partial z/\partial y + \partial x/\partial z) = x^2(0+0+0) = 0$$

$$(\nabla \phi) \cdot \mathbf{F} = (2x\mathbf{i}) \cdot (y\mathbf{i} + z\mathbf{j} + x\mathbf{k}) = 2xy + 0 + 0 = 2xy$$

$$\text{RHS} = 0 + 2xy = 2xy \quad \checkmark$$

FINAL ANSWER

$$\text{LHS} = \text{RHS} = 2xy. \text{ Identity verified! } \checkmark$$

Example 24 – Laplacian: Find $\nabla^2\phi$ for $\phi = x^3 + y^3 + z^3 - 3xyz$

STEP 2: FORMULA

$$\nabla^2\phi = \partial^2\phi/\partial x^2 + \partial^2\phi/\partial y^2 + \partial^2\phi/\partial z^2$$

STEP 5: SECOND DERIVATIVES

$$\partial\phi/\partial x = 3x^2 - 3yz \rightarrow \partial^2\phi/\partial x^2 = 6x$$

$$\partial\phi/\partial y = 3y^2 - 3xz \rightarrow \partial^2\phi/\partial y^2 = 6y$$

$$\partial\phi/\partial z = 3z^2 - 3xy \rightarrow \partial^2\phi/\partial z^2 = 6z$$

FINAL ANSWER

$$\nabla^2\phi = 6x + 6y + 6z = 6(x + y + z)$$

Example 25 – Find Divergence of $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ (position vector)

STEP 5

$$\partial x/\partial x + \partial y/\partial y + \partial z/\partial z = 1 + 1 + 1 = 3$$

FINAL ANSWER

$$\nabla \cdot \mathbf{r} = 3 \text{ (always, for any point in 3D space)}$$

Example 26 – Find Curl of $\mathbf{r}$ (position vector)

STEP 5: $\mathbf{R} = X\mathbf{I} + Y\mathbf{J} + Z\mathbf{K}$, $F_1 = X$, $F_2 = Y$, $F_3 = Z$

$$\text{i: } \partial z/\partial y - \partial y/\partial z = 0 - 0 = 0$$

$$\text{j: } -(\partial z/\partial x - \partial x/\partial z) = -(0 - 0) = 0$$

$$\text{k: } \partial y/\partial x - \partial x/\partial y = 0 - 0 = 0$$

FINAL ANSWER

$$\nabla \times \mathbf{r} = \mathbf{0} \text{ (position vector field is irrotational)}$$

Example 27 – Directional Derivative of $\phi = xy+yz+zx$ in direction $(1,2,2)$ at $(1,2,0)$

STEP 5: GRADIENT

$$\partial\phi/\partial x = y+z, \partial\phi/\partial y = x+z, \partial\phi/\partial z = y+x$$

$$\text{At } (1,2,0): \nabla\phi = 2\mathbf{i}+1\mathbf{j}+3\mathbf{k}$$

STEP 7: UNIT VECTOR

$$\mathbf{a}=(1,2,2), |\mathbf{a}|=\sqrt{1+4+4}=3, \hat{\mathbf{a}} = (\mathbf{i}+2\mathbf{j}+2\mathbf{k})/3$$

STEP 8: DOT PRODUCT

$$D = (2\mathbf{i}+\mathbf{j}+3\mathbf{k}) \cdot (\mathbf{i}+2\mathbf{j}+2\mathbf{k})/3 = (2+2+6)/3 = 10/3$$

FINAL ANSWER

$$\text{Directional Derivative} = 10/3$$

Example 28 – For $\mathbf{F} = ax\mathbf{i}+by\mathbf{j}+cz\mathbf{k}$, find conditions for solenoidal

STEP 5

$$\nabla \cdot \mathbf{F} = \partial(ax)/\partial x + \partial(by)/\partial y + \partial(cz)/\partial z = a+b+c$$

FINAL ANSWER

$$\text{Condition: } \mathbf{a+b+c = 0}$$

Example 29 – Curl of $\mathbf{F} = (y-z)\mathbf{i} + (z-x)\mathbf{j} + (x-y)\mathbf{k}$

STEP 1

$$F_1=y-z, F_2=z-x, F_3=x-y$$

STEP 5

$$\mathbf{i}: \partial(x-y)/\partial y - \partial(z-x)/\partial z = -1-1 = -2$$

$$\mathbf{j}: -[\partial(x-y)/\partial x - \partial(y-z)/\partial z] = -[1-(-1)] = -2$$

$$\mathbf{k}: \partial(z-x)/\partial x - \partial(y-z)/\partial y = -1-1 = -2$$

FINAL ANSWER

$$\nabla \times \mathbf{F} = -2\mathbf{i}-2\mathbf{j}-2\mathbf{k} = -2(\mathbf{i}+\mathbf{j}+\mathbf{k})$$

Example 30 – Find $\nabla(1/r)$ and show it equals $-\mathbf{r}/r^3$

STEP 5

$$1/r = (x^2 + y^2 + z^2)^{-1/2}$$

$$\partial(1/r)/\partial x = -(1/2) \cdot 2x \cdot (x^2 + y^2 + z^2)^{-3/2} = -x/r^3$$

$$\text{Similarly: } \partial(1/r)/\partial y = -y/r^3, \partial(1/r)/\partial z = -z/r^3$$

FINAL ANSWER

$$\nabla(1/r) = -(\mathbf{x}\mathbf{i} + \mathbf{y}\mathbf{j} + \mathbf{z}\mathbf{k})/r^3 = -\mathbf{r}/r^3$$

$$\text{Also written as: } \nabla(r^{-1}) = -r^{-3} \cdot \mathbf{r} = -\hat{\mathbf{r}}/r^2 \quad \checkmark$$

Example 31 – Show $\mathbf{F} = 2xy\mathbf{i} + (x^2 - z^2)\mathbf{j} - 2yz\mathbf{k}$ is irrotational; find scalar potential

STEP 5: CURL COMPONENTS

$$\mathbf{i}: \partial(-2yz)/\partial y - \partial(x^2 - z^2)/\partial z = -2z - (-2z) = 0$$

$$\mathbf{j}: -[\partial(-2yz)/\partial x - \partial(2xy)/\partial z] = -[0 - 0] = 0$$

$$\mathbf{k}: \partial(x^2 - z^2)/\partial x - \partial(2xy)/\partial y = 2x - 2x = 0 \quad \checkmark$$

STEP 6: FIND POTENTIAL $\Phi = \int \mathbf{F} \cdot d\mathbf{r}$

$$\partial\Phi/\partial x = 2xy \rightarrow \Phi = x^2y + f(y, z)$$

$$\partial\Phi/\partial y = x^2 + \partial f/\partial y = x^2 - z^2 \rightarrow \partial f/\partial y = -z^2 \rightarrow f = -yz^2 + g(z)$$

$$\partial\Phi/\partial z = -2yz + g'(z) = -2yz \rightarrow g'(z) = 0 \rightarrow g = C$$

FINAL ANSWER

$$\Phi = x^2y - yz^2 + C$$

Example 32 – Verify: $\nabla(\phi\psi) = \phi\nabla\psi + \psi\nabla\phi$ for $\phi=x^2, \psi=y^2$

STEP 5: LHS

$$\phi\psi = x^2y^2, \nabla(x^2y^2) = 2xy^2\mathbf{i} + 2x^2y\mathbf{j} + 0\mathbf{k}$$

STEP 6: RHS

$$\phi\nabla\psi = x^2 \cdot (0\mathbf{i} + 2y\mathbf{j} + 0\mathbf{k}) = 2x^2y\mathbf{j}$$

$$\psi\nabla\phi = y^2 \cdot (2x\mathbf{i} + 0\mathbf{j} + 0\mathbf{k}) = 2xy^2\mathbf{i}$$

$$\text{Sum: } 2xy^2\mathbf{i} + 2x^2y\mathbf{j} \quad \checkmark$$

FINAL ANSWER

$$\text{LHS} = \text{RHS} = 2xy^2\mathbf{i} + 2x^2y\mathbf{j}. \text{ Identity verified! } \checkmark$$

Example 33 – Find gradient of $\phi = \sin(x+y) + \cos(y+z)$

STEP 5

$$\partial\phi/\partial x = \cos(x+y)$$

$$\partial\phi/\partial y = \cos(x+y) - \sin(y+z)$$

$$\partial\phi/\partial z = -\sin(y+z)$$

FINAL ANSWER

$$\nabla\phi = \cos(x+y)\mathbf{i} + [\cos(x+y) - \sin(y+z)]\mathbf{j} - \sin(y+z)\mathbf{k}$$

Example 34 – Divergence of $\mathbf{F} = x^2z\mathbf{i} - 2y^3z^2\mathbf{j} + xy^2z\mathbf{k}$ at $(1, -1, 1)$

STEP 5

$$\partial(x^2z)/\partial x = 2xz \rightarrow \text{at } (1, -1, 1): 2$$

$$\partial(-2y^3z^2)/\partial y = -6y^2z^2 \rightarrow \text{at } (1, -1, 1): -6$$

$$\partial(xy^2z)/\partial z = xy^2 \rightarrow \text{at } (1, -1, 1): 1$$

FINAL ANSWER

$$\nabla \cdot \mathbf{F} \text{ at } (1, -1, 1) = 2 - 6 + 1 = -3$$

Example 35 – Verify Curl of gradient of $\phi=xyz$ equals zero

STEP 5

$\nabla(xyz) = yzi + xzj + xyk$. Let $G_1=yz$, $G_2=xz$, $G_3=xy$

i: $\partial(xy)/\partial y - \partial(xz)/\partial z = x - x = 0$

j: $-\partial(xy)/\partial x - \partial(yz)/\partial z = -[y - y] = 0$

k: $\partial(xz)/\partial x - \partial(yz)/\partial y = z - z = 0$

FINAL ANSWER

$$\nabla \times (\nabla \phi) = \mathbf{0} \quad \checkmark$$

Example 36 – For what value of n is $\nabla \cdot (r^n \mathbf{r}) = 0$?

STEP 5: USE IDENTITY $\nabla \cdot (R^n \mathbf{R}) = (n+3)R^n$

$$\nabla \cdot (r^n \mathbf{r}) = r^n (\nabla \cdot \mathbf{r}) + \mathbf{r} \cdot \nabla (r^n) = r^n \cdot 3 + \mathbf{r} \cdot n \mathbf{r} r^{n-2} \mathbf{r} = 3r^n + nr^n = (n+3)r^n$$

STEP 6: SET TO ZERO

$$(n+3)r^n = 0 \rightarrow n = -3$$

FINAL ANSWER

$$n = -3$$

Example 37 – Find gradient of $\phi = 2x^2 - 3y^2 + 4z$ at $(1, 1, -1)$; find directional derivative toward $(2, 3, -1)$

STEP 5: GRADIENT

$$\partial\phi/\partial x = 4x, \partial\phi/\partial y = -6y, \partial\phi/\partial z = 4$$

$$\text{At } (1, 1, -1): \nabla\phi = 4\mathbf{i} - 6\mathbf{j} + 4\mathbf{k}$$

STEP 7: DIRECTION TOWARD $(2, 3, -1)$ FROM $(1, 1, -1)$

$$\mathbf{a} = (2-1)\mathbf{i} + (3-1)\mathbf{j} + (-1-(-1))\mathbf{k} = \mathbf{i} + 2\mathbf{j} + 0\mathbf{k}$$

$$|\mathbf{a}| = \sqrt{1+4} = \sqrt{5}$$

$$\hat{\mathbf{a}} = (\mathbf{i} + 2\mathbf{j})/\sqrt{5}$$

STEP 8

$$D = (4\mathbf{i} - 6\mathbf{j} + 4\mathbf{k}) \cdot (\mathbf{i} + 2\mathbf{j})/\sqrt{5} = (4 - 12)/\sqrt{5} = -8/\sqrt{5} = -8\sqrt{5}/5$$

FINAL ANSWER

$$\text{Directional Derivative} = -8/\sqrt{5}$$

Example 38 – Prove $\nabla \times (\nabla \times \mathbf{F}) = \nabla(\nabla \cdot \mathbf{F}) - \nabla^2 \mathbf{F}$

STEP 2: USE VECTOR IDENTITY APPROACH (X-COMPONENT)

x-component of $\nabla \times (\nabla \times \mathbf{F})$:

$$= \partial/\partial y (\partial F_1/\partial y - \partial F_2/\partial x) - \partial/\partial z (\partial F_3/\partial x - \partial F_1/\partial z)$$

$$= \partial^2 F_1/\partial y^2 - \partial^2 F_2/\partial x \partial y - \partial^2 F_3/\partial z \partial x + \partial^2 F_1/\partial z^2$$

Add and subtract $\partial^2 F_1/\partial x^2$:

$$= (\partial^2 F_1/\partial x^2 + \partial^2 F_1/\partial y^2 + \partial^2 F_1/\partial z^2) \text{ rewritten + cross terms}$$

$$= \partial/\partial x (\partial F_1/\partial x + \partial F_2/\partial y + \partial F_3/\partial z) - \nabla^2 F_1$$

$$= \partial(\nabla \cdot \mathbf{F})/\partial x - \nabla^2 F_1 \quad \checkmark \text{ (x-component of identity)}$$

Similarly for y, z components $\rightarrow$ Identity proved.

FINAL ANSWER

$$\nabla \times (\nabla \times \mathbf{F}) = \nabla(\nabla \cdot \mathbf{F}) - \nabla^2 \mathbf{F} \quad \checkmark$$

Example 39 – If $\phi = x^2 + y^2 - z^2$, find $\nabla^2\phi$ and state if it satisfies Laplace's equation

STEP 5: SECOND DERIVATIVES

$$\partial^2\phi/\partial x^2 = 2, \partial^2\phi/\partial y^2 = 2, \partial^2\phi/\partial z^2 = -2$$

STEP 6: LAPLACIAN

$$\nabla^2\phi = 2+2-2 = 2 \neq 0$$

FINAL ANSWER

$\nabla^2\phi = 2$. Does NOT satisfy Laplace's equation ($\nabla^2\phi = 0$ required).

Example 40 – Find Curl of $F = (x^2 + y^2 + z^2)(xi + yj + zk)$

STEP 1

$$F = r^2 \cdot r \text{ where } r^2 = x^2 + y^2 + z^2, r = xi + yj + zk$$

$$F_1 = (x^2 + y^2 + z^2)x, F_2 = (x^2 + y^2 + z^2)y, F_3 = (x^2 + y^2 + z^2)z$$

STEP 5: I-COMPONENT OF CURL

$$\partial F_3/\partial y - \partial F_2/\partial z = \partial[(x^2 + y^2 + z^2)z]/\partial y - \partial[(x^2 + y^2 + z^2)y]/\partial z$$

$$= 2yz - 2yz = 0$$

By symmetry, all components are 0.

FINAL ANSWER

$\nabla \times F = \mathbf{0}$. The field $r^2 \cdot r$ is irrotational! This is because $F = \nabla(r^4/4)$ — it's a gradient field.

Example 41 – Tangent Plane to $x^2 + 2y^2 + 3z^2 = 6$ at $(1, 1, 1)$

STEP 5

$$\phi = x^2 + 2y^2 + 3z^2, \nabla\phi = 2xi + 4yj + 6zk \rightarrow \text{at } (1, 1, 1): 2i + 4j + 6k$$

STEP 6: TANGENT PLANE EQUATION

$$2(x-1) + 4(y-1) + 6(z-1) = 0 \rightarrow 2x + 4y + 6z = 12 \rightarrow x + 2y + 3z = 6$$

FINAL ANSWER

Tangent Plane: $x + 2y + 3z = 6$

Example 42 – Find maximum value of directional derivative of $\phi = x^3y^2z$ at $(1,1,1)$

STEP 5

$$\partial\phi/\partial x = 3x^2y^2z \rightarrow 3, \partial\phi/\partial y = 2x^3yz \rightarrow 2, \partial\phi/\partial z = x^3y^2 \rightarrow 1 \text{ at } (1,1,1)$$

$$\nabla\phi = 3\mathbf{i} + 2\mathbf{j} + \mathbf{k}$$

STEP 6: MAXIMUM = $|\nabla\phi|$

$$|\nabla\phi| = \sqrt{9+4+1} = \sqrt{14}$$

FINAL ANSWER

Maximum Directional Derivative = $\sqrt{14}$ in direction $(3\mathbf{i} + 2\mathbf{j} + \mathbf{k})/\sqrt{14}$

Example 43 – Prove $\nabla \cdot (\mathbf{F} \times \mathbf{G}) = \mathbf{G} \cdot (\nabla \times \mathbf{F}) - \mathbf{F} \cdot (\nabla \times \mathbf{G})$

STEP 2: EXPAND LHS

$$\mathbf{F} \times \mathbf{G} = (F_2G_3 - F_3G_2)\mathbf{i} - (F_1G_3 - F_3G_1)\mathbf{j} + (F_1G_2 - F_2G_1)\mathbf{k}$$

$$\nabla \cdot (\mathbf{F} \times \mathbf{G}) = \partial(F_2G_3 - F_3G_2)/\partial x + \partial(F_1G_3 - F_3G_1)/\partial y + \dots \text{ (full expansion)}$$

After applying product rule and rearranging all terms (each with 2 terms, 6 total):

$$= \mathbf{G} \cdot (\nabla \times \mathbf{F}) - \mathbf{F} \cdot (\nabla \times \mathbf{G}) \quad \checkmark \text{ (by rearranging the 6-term expansion)}$$

FINAL ANSWER

$$\nabla \cdot (\mathbf{F} \times \mathbf{G}) = \mathbf{G} \cdot (\nabla \times \mathbf{F}) - \mathbf{F} \cdot (\nabla \times \mathbf{G}) \quad \checkmark$$

Example 44 – Determine if $\mathbf{F} = (y+z)\mathbf{i} + (z+x)\mathbf{j} + (x+y)\mathbf{k}$ is solenoidal and/or irrotational

STEP 5A: DIVERGENCE

$$\partial(y+z)/\partial x + \partial(z+x)/\partial y + \partial(x+y)/\partial z = 0+0+0 = 0 \rightarrow \text{Solenoidal } \checkmark$$

STEP 5B: CURL

$$\mathbf{i}: \partial(x+y)/\partial y - \partial(z+x)/\partial z = 1-1 = 0$$

$$\mathbf{j}: -[\partial(x+y)/\partial x - \partial(y+z)/\partial z] = -[1-1] = 0$$

$$\mathbf{k}: \partial(z+x)/\partial x - \partial(y+z)/\partial y = 1-1 = 0 \rightarrow \text{Irrotational } \checkmark$$

FINAL ANSWER

F is BOTH Solenoidal and Irrotational!

Example 45 – Full problem: $\mathbf{F} = (6xy+z^3)\mathbf{i} + (3x^2-z)\mathbf{j} + (3xz^2-y)\mathbf{k}$. Check irrotational; find φ

STEP 1

$$F_1=6xy+z^3, F_2=3x^2-z, F_3=3xz^2-y$$

STEP 5: CURL CHECK

$$\text{i: } \partial(3xz^2-y)/\partial y - \partial(3x^2-z)/\partial z = -1 - (-1) = 0$$

$$\text{j: } -[\partial(3xz^2-y)/\partial x - \partial(6xy+z^3)/\partial z] = -[3z^2-3z^2] = 0$$

$$\text{k: } \partial(3x^2-z)/\partial x - \partial(6xy+z^3)/\partial y = 6x-6x = 0 \quad \checkmark \text{ Irrotational}$$

STEP 6: FIND SCALAR POTENTIAL

$$\partial\varphi/\partial x = 6xy+z^3 \rightarrow \varphi = 3x^2y + xz^3 + f(y,z)$$

$$\partial\varphi/\partial y = 3x^2 + \partial f/\partial y = 3x^2-z \rightarrow \partial f/\partial y = -z \rightarrow f = -yz+g(z)$$

$$\partial\varphi/\partial z = 3xz^2 + (-y) + g'(z) = 3xz^2-y \rightarrow g'(z)=0 \rightarrow g=C$$

FINAL ANSWER

$$\varphi = 3x^2y + xz^3 - yz + C$$

Section 16 – Exercise Problems (50 Questions)

Part A – 2-Mark Questions (Definitions and Basic Operations)

1. Define a scalar point function with two examples.

2M

2. Define a vector point function with two examples.

2M

3. Write the del operator ∇ in expanded form.

2M

4. Define gradient of a scalar function.

2M

5. Define divergence of a vector field and state what it represents physically.

2M

6. Define curl of a vector field.

2M

7. When is a vector field called solenoidal?

2M

8. When is a vector field called irrotational?

2M

9. Find $\nabla\phi$ for $\phi = x^2 + y^2 + z^2$.

2M

10. Find $\nabla \cdot \mathbf{F}$ for $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$.

2M

11. Find $\nabla \times \mathbf{r}$ where $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$.

2M

12. State the formula for directional derivative.

2M

13. What is the physical meaning of gradient?

2M

14. State the vector identity $\nabla \times (\nabla \phi) = ?$

2M

15. State the vector identity $\nabla \cdot (\nabla \times \mathbf{F}) = ?$

2M

16. Find $\nabla(r^n)$ in terms of r and n .

2M

17. Find the unit normal to the surface $x + 2y + 3z = 6$.

2M

18. What is the Laplace operator ∇^2 ?

2M

19. Find $\nabla \cdot \mathbf{F}$ where $\mathbf{F} = x^2\mathbf{i} + y^2\mathbf{j} + z^2\mathbf{k}$.

2M

20. State the maximum directional derivative theorem.

2M

Part B – 5-Mark Questions (Full Vector Operations)

21. Find $\nabla\phi$ for $\phi = x^3y^2z$ and evaluate at $(1, 2, -1)$.

5M

22. Find the directional derivative of $\phi = xy + yz + zx$ at $(1, 1, 1)$ in the direction of $\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$.

5M

23. Find unit normal vector to the surface $x^2y+2xz^2=8$ at $(1,2,1)$.

5M

24. Show that $F = (y^2z^3)\mathbf{i} + (2xyz^3)\mathbf{j} + (3xy^2z^2)\mathbf{k}$ is irrotational and find scalar potential.

5M

25. Find $\nabla \cdot F$ and $\nabla \times F$ for $F = (xy^2)\mathbf{i} + (y^3z)\mathbf{j} + (xz^2)\mathbf{k}$.

5M

26. Prove $\nabla \cdot (\phi F) = \phi(\nabla \cdot F) + (\nabla \phi) \cdot F$ using specific functions $\phi=xy$, $F=xy\mathbf{i}+yz\mathbf{j}+zx\mathbf{k}$.

5M

27. Show $F = (2xy+z)\mathbf{i} + (x^2-2yz)\mathbf{j} + (x-y^2)\mathbf{k}$ is irrotational. Find scalar potential.

5M

28. Find the equation of tangent plane to $x^2+y^2+z^2=14$ at $(1,2,3)$.

5M

29. If $\phi = x^2y^2z^2$, find $\nabla^2\phi$.

5M

30. Prove $\nabla(\phi+\psi) = \nabla\phi + \nabla\psi$ from first principles.

5M

31. Find angle between surfaces $x^2+y^2+z^2=9$ and $z=x^2+y^2-3$ at $(2,-1,2)$.

5M

32. Show $F = (y+z-x)\mathbf{i} + (z+x-y)\mathbf{j} + (x+y-z)\mathbf{k}$ is solenoidal.

5M

33. Find $\nabla \times F$ for $F = xy^2z\mathbf{i} + x^2yz\mathbf{j} + xyz^2\mathbf{k}$ and find $\nabla \cdot (\nabla \times F)$.

5M

34. For $F = (x^2-yz)\mathbf{i} + (y^2-zx)\mathbf{j} + (z^2-xy)\mathbf{k}$, find $\text{div } F$ and $\text{curl } F$.

5M

35. Prove $\nabla \cdot (r/r^n) = (3-n)/r^n$.

5M

36. Find max directional derivative of $\phi=x^2yz^3$ at $(2,1,-1)$.

5M

37. Find a,b,c if $F = (ax+3y+z)\mathbf{i} + (x-2y+bz)\mathbf{j} + (cx+y+3z)\mathbf{k}$ is irrotational.

5M

38. Show $\nabla^2(r^n) = n(n+1)r^{n-2}$.

5M

39. If $\nabla\phi = \mathbf{0}$ throughout a region, what can you conclude about ϕ ? Explain.

5M

40. Verify $\nabla \times (\phi F) = \phi(\nabla \times F) + (\nabla \phi) \times F$ for $\phi=x^2$, $F=y\mathbf{i}+z\mathbf{j}+x\mathbf{k}$.

5M

41. Find the directional derivative of $\phi=4e^{(2x-y+z)}$ at $(1,1,-1)$ in direction toward $(-3,5,6)$.

5M

42. Prove that if F is both solenoidal and irrotational, then $\nabla^2 F = \mathbf{0}$.

5M

43. Show $\nabla(1/r) = -r/r^3$ and hence $\nabla \cdot (1/r)$ has no meaning. What is $\nabla \cdot \nabla(1/r)$?

5M

44. $F = \nabla(x^3+y^3+z^3-3xyz)$. Find $\text{curl } F$ and verify it is solenoidal.

5M

45. Find the scalar potential of $F = yz\mathbf{i} + zx\mathbf{j} + xy\mathbf{k}$ (if it exists). Verify.

5M

46. Prove: $\nabla \cdot (F \times G) = G \cdot (\nabla \times F) - F \cdot (\nabla \times G)$ using $F=x^2\mathbf{i}+y\mathbf{j}$, $G=y\mathbf{i}+x^2\mathbf{j}$.

5M

47. If $r = x\mathbf{i}+y\mathbf{j}+z\mathbf{k}$ and $f(r)$ is any scalar function of r , prove $\nabla \times (f(r)r) = \mathbf{0}$.

5M

48. Find $\nabla^2(\log r)$ and determine for what condition it is zero.

5M

49. Show that $\nabla(\mathbf{r} \cdot \mathbf{a}) = \mathbf{a}$ where $\mathbf{a}$ is a constant vector and $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$.

5M

50. $\mathbf{F} = (x+2y+az)\mathbf{i} + (bx-3y-z)\mathbf{j} + (4x+cy+2z)\mathbf{k}$. Find a, b, c for $\mathbf{F}$ to be irrotational. Then find ϕ .

5M

Theorems, Formula Sheet & Quick Revision

Key Theorems

Theorem 1 – Gradient Theorem: The gradient of a scalar function at a point gives the direction and rate of maximum increase of the function. The maximum directional derivative equals $|\nabla\phi|$.

Theorem 2 – Normal Vector Theorem: The gradient $\nabla\phi$ is always perpendicular (normal) to the level surface $\phi = c$ at any point on the surface.

Theorem 3 – Irrotational Field = Conservative Field: A vector field F is irrotational ($\nabla\times F = 0$) if and only if it can be expressed as the gradient of some scalar potential ϕ (i.e., $F = \nabla\phi$).

Theorem 4 – Curl of Gradient = Zero: For any scalar function ϕ : $\nabla\times(\nabla\phi) = 0$. (Gradient fields have no rotation.)

Theorem 5 – Divergence of Curl = Zero: For any vector field F : $\nabla\cdot(\nabla\times F) = 0$. (Curl fields have no sources/sinks.)

Complete Formula Sheet

Del Operator

∇ (Del): $i(\partial/\partial x) + j(\partial/\partial y) + k(\partial/\partial z)$

Gradient

$\text{grad } \phi = \nabla\phi$: $(\partial\phi/\partial x)i + (\partial\phi/\partial y)j + (\partial\phi/\partial z)k$

$\nabla(\phi+\psi)$: $\nabla\phi + \nabla\psi$

$\nabla(\phi\psi)$: $\phi\nabla\psi + \psi\nabla\phi$

$\nabla(r^n)$: $n \cdot r^{n-2} \cdot r$

∇r : $r/r = \hat{r}$

$\nabla(1/r)$: $-r/r^3$

Divergence

$\text{div } F = \nabla\cdot F$: $\partial F_1/\partial x + \partial F_2/\partial y + \partial F_3/\partial z$

$\nabla\cdot r$: 3

$\nabla\cdot(r/r^n)$: $(3-n)/r^n$

$\nabla\cdot(r^n r)$: $(n+3)r^n$

$\nabla\cdot(\phi F)$: $\phi(\nabla\cdot F) + (\nabla\phi)\cdot F$

Curl

$\text{curl } F = \nabla\times F$: $i(\partial F_3/\partial y - \partial F_2/\partial z) - j(\partial F_3/\partial x - \partial F_1/\partial z) + k(\partial F_2/\partial x - \partial F_1/\partial y)$

$\nabla\times r$: 0

$\nabla\times(\phi F)$: $\phi(\nabla\times F) + (\nabla\phi)\times F$

$\nabla \times (\nabla \phi)$: 0 (always)

Directional Derivative

$D_{\mathbf{a}} \phi$: $\nabla \phi \cdot \hat{\mathbf{a}}$ where $\hat{\mathbf{a}} = \mathbf{a}/|\mathbf{a}|$

Max directional deriv: $|\nabla \phi|$ (in direction $\nabla \phi$)

Laplacian

$\nabla^2 \phi$: $\partial^2 \phi / \partial x^2 + \partial^2 \phi / \partial y^2 + \partial^2 \phi / \partial z^2$

$\nabla^2(\mathbf{r}^n)$: $n(n+1) \mathbf{r}^{n-2}$

$\nabla^2(\log r)$: $1/r^2$ (in 3D, actually $(3-2)/r^2 = 1/r^2$ – use $\nabla^2(\log r)$ for 2D)

Unit Normal & Tangent Plane

Unit Normal $\hat{\mathbf{n}}$: $\nabla \phi / |\nabla \phi|$

Tangent Plane at $\mathbf{P}_0$: $(\nabla \phi)_0 \cdot (\mathbf{r} - \mathbf{r}_0) = 0$

Angle between surfaces: $\cos \theta = (\nabla \phi_1 \cdot \nabla \phi_2) / (|\nabla \phi_1| |\nabla \phi_2|)$

Important Vector Identities

$\nabla \cdot (\nabla \times \mathbf{F})$: 0

$\nabla \times (\nabla \phi)$: 0

$\nabla \times (\nabla \times \mathbf{F})$: $\nabla(\nabla \cdot \mathbf{F}) - \nabla^2 \mathbf{F}$

$\nabla \cdot (\mathbf{F} \times \mathbf{G})$: $\mathbf{G} \cdot (\nabla \times \mathbf{F}) - \mathbf{F} \cdot (\nabla \times \mathbf{G})$

$\nabla \times (\mathbf{F} \times \mathbf{G})$: $\mathbf{F}(\nabla \cdot \mathbf{G}) - \mathbf{G}(\nabla \cdot \mathbf{F}) + (\mathbf{G} \cdot \nabla)\mathbf{F} - (\mathbf{F} \cdot \nabla)\mathbf{G}$

Quick Revision Notes

KEY FORMULAS TO MEMORIZE

- $\nabla = \mathbf{i}\partial/\partial x + \mathbf{j}\partial/\partial y + \mathbf{k}\partial/\partial z$
- Gradient: $\nabla \phi \rightarrow$ apply ∇ to scalar $\rightarrow$ get vector
- Divergence: $\nabla \cdot \mathbf{F} \rightarrow$ dot product $\rightarrow$ get scalar
- Curl: $\nabla \times \mathbf{F} \rightarrow$ cross product $\rightarrow$ get vector
- $\nabla r = \mathbf{r}/r$, $\nabla \times \mathbf{r} = \mathbf{0}$, $\nabla(r^n) = nr^{n-2}\mathbf{r}$
- Max dir deriv = $|\nabla \phi|$; direction = $\nabla \phi / |\nabla \phi|$

COMMON EXAM MISTAKES — AVOID THESE!

- ✗ Forgetting the NEGATIVE sign on the j-component in curl expansion
- ✗ Using a non-unit vector for directional derivative (ALWAYS divide by $|\mathbf{a}|$ first!)
- ✗ Taking $\partial F_1 / \partial y$ instead of $\partial F_1 / \partial x$ for divergence
- ✗ Getting gradient wrong by treating all variables as x (must vary one at a time!)
- ✗ Forgetting that divergence gives scalar but curl gives vector
- ✗ Forgetting to verify the given point lies on the surface before finding normal

SHORTCUT INSIGHTS (After mastering the full method)

- For $\mathbf{F} = f(r)\mathbf{r}$ type fields: $\nabla \times \mathbf{F} = \mathbf{0}$ always (always irrotational)
- For linear fields $\mathbf{F} = (ax+by+cz)\mathbf{i} + \dots$: curl has only constant terms
- If $\nabla \cdot \mathbf{F} = 0$ and $\nabla \times \mathbf{F} = \mathbf{0}$, then $\nabla^2 \mathbf{F} = \mathbf{0}$ (harmonic vector field)

- $\nabla^2(1/r) = 0$ everywhere except origin ($r \neq 0$)
- $\nabla \cdot \mathbf{r} = 3$ always — memorize this!
- To check if $\mathbf{F} = \nabla\phi$ is possible: verify $\nabla \times \mathbf{F} = \mathbf{0}$ first, then integrate

Summary Table — Properties Comparison

Operation	Input	Output	Formula	Special Name
Gradient	Scalar ϕ	Vector	$\nabla\phi$	Points in max increase direction
Divergence	Vector $\mathbf{F}$	Scalar	$\nabla \cdot \mathbf{F}$	Solenoidal if $\nabla \cdot \mathbf{F} = 0$
Curl	Vector $\mathbf{F}$	Vector	$\nabla \times \mathbf{F}$	Irrotational if $\nabla \times \mathbf{F} = \mathbf{0}$
Laplacian	Scalar ϕ	Scalar	$\nabla^2\phi = \nabla \cdot (\nabla\phi)$	Harmonic if $\nabla^2\phi = 0$
Dir. Deriv.	Scalar ϕ	Scalar	$\nabla\phi \cdot \hat{\mathbf{a}}$	Max = $ \nabla\phi $

Important Results for Special Functions

Function	∇f	Remarks
$f = c$ (const)	$\mathbf{0}$	Gradient of constant is zero
$f = r$	$\hat{\mathbf{r}} = \mathbf{r}/r$	Unit radial vector
$f = r^2$	$2\mathbf{r}$	Radially outward
$f = r^n$	$n r^{n-2} \mathbf{r}$	General power formula
$f = 1/r$	$-\mathbf{r}/r^3 = -\hat{\mathbf{r}}/r^2$	Radially inward (like gravity)
$f = \log r$	$\mathbf{r}/r^2 = \hat{\mathbf{r}}/r$	Slowly decreasing outward

END OF UNIT IV — VECTOR DIFFERENTIATION

This module covers: Scalar & Vector Functions · Del Operator · Gradient · Directional Derivative · Unit Normal · Divergence · Curl · Vector Identities · 45 Solved Examples · 50 Exercise Problems · Complete Formula Sheet

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